

LESSON 1: MACHINE LEARNING

One of the fundamental attributes of an intelligent system is **learning**—the ability to acquire and use information. Learning involves creating internal links, or **associations**, between what is already known and new information. At its core, learning is the capacity to form new associations and to make use of them.

LEARNING IS THE ABILITY TO FORM AND USE NEW ASSOCIATIONS.

EXAMPLE 1.1

Consider a toddler. Through experience, she recognizes the shape of a cup and understands that it contains water to quench her thirst. These are associations already present in her mind: <cup as an object>, <water>, <thirst>. She learned them through repeated experience.

Now something new happens. Maya feels thirsty and points to the cup. Her mother says, "cup" and hands Maya the cup. The next day, the same event repeats: Maya points to the cup, her mother says, "cup," and gives it to her. After a few days of this repeating pattern, Maya feels thirsty but does not see the cup nearby. This time, she looks at her mother and says, "cup."



Through repetition, Maya recognized a pattern: whenever she heard the word 'cup', she received the cup itself. This formed an association in her mind between the word 'cup' and the object. She learned that the object's name is 'cup'.

Maya has learned. She created a new association in her mind: linking the object cup to the word ‘cup’. This learning occurred because she recognized a repeating pattern—the sound "cup" was always paired with the actual cup—and reinforced it through experience.

From this simple example, we can identify three fundamental attributes of learning:

1. It requires the creation of links or **associations** between things.
2. It depends on the ability to recognize **patterns**.
3. It usually requires **repetition** to strengthen those links.

MACHINE LEARNING

Artificial Intelligence (AI) is the field of computer science that focuses on building machines that can perform tasks that usually require human intelligence.

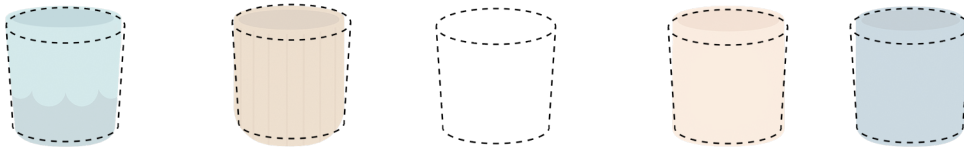
One branch of AI is called machine learning—this is when a computer is designed to have learning capabilities like Maya’s. Through repetition, the computer can identify patterns in data and form internal associations in memory. Once these associations are established and reinforced, the computer can make use of them.

EXAMPLE 1.2

An AI system can also be taught to identify a cup. To do this, it is given an image of a cup and is instructed to associate that image with the word ‘cup’. But to ensure it can recognize cups of all kinds, the computer is shown many images of cups in different sizes, colors, and forms.



By reviewing these images repeatedly, the computer begins to identify the general shape that defines a cup. This ability is called pattern recognition—the capacity to detect recurring regularities, in this case, the overall shape of a cup.



Shared Shape

These images show cups of different sizes, colors, and details. By seeing many different examples, the computer begins to identify the shared shape that makes an object a cup, even when the style, color, or design changes. The ability to identify this shared shape is called pattern recognition.

PATTERN RECOGNITION IS THE ABILITY TO DETECT RECURRING REGULARITIES.

TYPES OF LEARNING

AI systems, like humans, can learn in different ways. There are two main types of machine learning: supervised learning and unsupervised learning.

Supervised learning is like teaching. Maya was given a cup and told, “This is a cup.” This happened repeatedly until she learned to recognize the shape of a cup and associate it with the word ‘cup’.

For an AI system, supervised learning works in a similar way. The computer is given data that already has labels and must find the pattern that links the data to those labels. For example, just as in the previous case, the AI is shown many images of cups, each labeled ‘cup,’ and it learns to associate the recurring shape with the label.

On the other hand, **unsupervised learning** is more like exploring. The AI is not told what to look for. Instead, it analyzes the data, identifies patterns, and groups similar items together based on shared features. Labels are not used during learning, but they can be added afterward to name each group once the patterns are understood.

EXAMPLE 1.3

Imagine Maya walking around the house each day and grouping objects by similarity: cups in one pile, shoes in another, crayons in another. Later, someone tells her the names of these groups: “cups,” “shoes,” and “crayons.” In this way, she would have learned without being told the names in advance.



In unsupervised learning, similar objects are grouped together without being told their names. Later, labels such as ‘cups,’ ‘shoes,’ and ‘crayons’ can be added to each group.

In a similar way, an AI system may be instructed to review hundreds of unlabeled images of household objects and group them based on recurring patterns of shape, color, or design. In such a case, it is said that the AI has learned unsupervised. The labels can be added to those groups afterward.

TWO COMMON TYPES OF LEARNING ARE SUPERVISED LEARNING
AND UNSUPERVISED LEARNING.

BUILDING THE MODEL

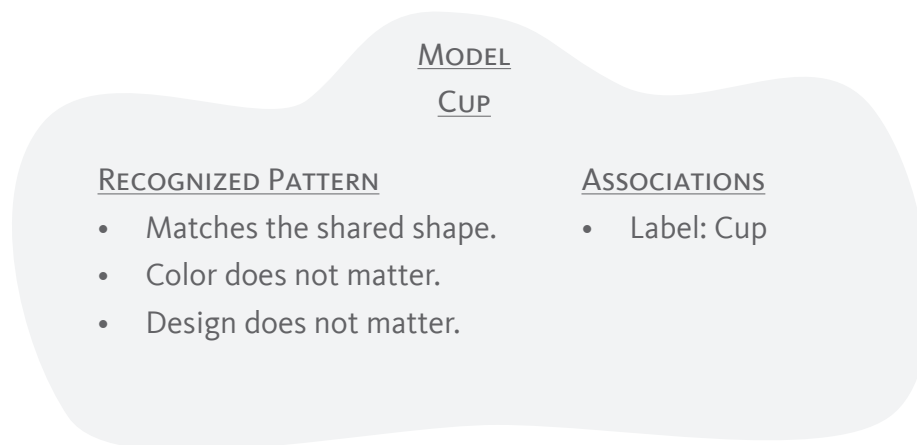
The process of learning results in the creation of a model. A model is the AI's internal representation of what it has learned — made up of the patterns it has recognized, the labels associated with those patterns, and the rules it has formed about which features are important and which are not.

A **MODEL** IS THE INTERNAL SYSTEM THAT REPRESENTS RECOGNIZED PATTERNS AND THE ASSOCIATIONS FORMED FROM THEM.

EXAMPLE 1.4

In this example, we will explore what would be contained in the AI's model of a cup. The most important element is the shared shape—the recurring pattern the AI has learned to identify. But the model also includes refinements, or rules, about features that do not affect whether something is a cup.

For instance, the style, color, or decorative designs are all noted but explicitly



This diagram shows a model of a cup. On the left are the recognized patterns that define a cup. The model identifies the shared shape as essential while explicitly instructing the AI that other variations—such as style, color, or design—do not matter for recognizing a cup. On the right are the associations linked to this pattern, namely the label 'cup'.

treated as irrelevant. As long as the object fits the shared shape, the model recognizes it as a cup.

Finally, the model includes the associations connected to this pattern, such as the label ‘cup’. This enables the computer to generalize, applying what it has learned to correctly identify cups it has never seen before.

TRAINING THE MODEL

Once a model is built, it must be tested to evaluate how well it performs. This testing is done on new data—examples the AI has not seen before. The goal is to determine whether the model can generalize what it has learned and accurately classify new inputs, such as recognizing a cup in an unfamiliar image.

Testing does not happen in isolation. As part of this process, the model’s predictions are compared to the correct answers. When it makes errors—known as misclassifications—those mistakes are analyzed. The model is then refined by adjusting how it interprets patterns and associations.

This ongoing process of learning from mistakes and improving performance by updating the model is known as training.

TRAINING IS THE PROCESS OF IMPROVING A MODEL USING NEW DATA AND CORRECTIONS.

Notes

While there is an official training phase that occurs after a model is first built and before it is released for use, training is not limited to that stage. In reality, training—or more accurately, refinement—is an ongoing process.

EXAMPLE 1.5

Now that the model for a cup has been built, it is tested on new images the AI has not seen before. The AI attempts to classify each image using the model it has created for 'cup'. It checks for the recurring pattern it has learned, disregards irrelevant features like color and design, and predicts which images contain cups based on that pattern.

The model correctly classified some images, but misclassified others. Upon review, it was found that the cups the model missed were those that had handles. The AI was informed that these examples were also cups and had been incorrectly excluded. It was then instructed to review those images, compare the patterns, and refine its model to correct the error.



Correctly Classified



Misclassified

This image shows how the model performed when tested on new data. On the left are cups that the model correctly classified based on the learned pattern. On the right are cups that were misclassified—these were not recognized because the model had not yet learned to accept handles as part of the pattern for 'cup'.

After reanalyzing the images, the model identified a new feature—the handle—and recognized that some cups have handles while others do not. It updated its internal representation accordingly, forming a new rule: “Handles do not matter.”

MODELCUPRECOGNIZED PATTERN

- Matches the shared shape.
- Color does not matter.
- Design does not matter.
- Include handles.

ASSOCIATIONS

- Label: Cup

After training, the model is updated to include cups with handles. It adds a rule to include handles as part of the recognized pattern, allowing the model to classify as ‘cup’ any object that matches the shared shape—whether it has a handle or not.

Now that the initial phase of training is complete, the model is ready for use. But learning does not stop here. While there is an official training phase that takes place after a model is first built and before it is released, training is not limited to that stage.

In practice, learning—or more precisely, refinement—is an ongoing process. As the model encounters new data, makes mistakes, or operates in changing conditions, it can continue to be adjusted and improved.

LEARNING IS A CONTINUOUS PROCESS.

Notes

Remind students: Learning doesn’t stop after the test — it’s a continuous process.

Just like an AI model, we don’t get everything right the first time. We learn through practice, mistakes, feedback, and adjustment. The more we apply what we’ve learned and reflect on our errors, the stronger our internal models become.

SUMMARY

Learning – the ability to form and use new associations.

- The three fundamental attributes of learning:
 1. the ability to recognize patterns.
 2. the ability to form associations.
 3. the need for repetition.
- **Pattern recognition** – the ability to detect recurring regularities.
- Two common types of learning are **supervised** (with labels) and **unsupervised** (without labels).
- **Model** – the internal system that represents recognized patterns and the associations formed from them.
- **Training** – the process of improving a model using new data and corrections.
- Learning is a continuous process.

Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Learning is the ability to form and use new associations.
<u>F</u>	Pattern recognition allows a system to detect and respond to random or unrelated features.
<u>T</u>	Supervised learning uses labeled data.
<u>F</u>	Unsupervised learning means the system receives labels in advance and uses them to form associations.
<u>T</u>	A model is the internal system that holds the patterns and associations a computer has learned.
<u>F</u>	Once a model is created, it can no longer be changed or improved.
<u>T</u>	Training is the process of improving a model by using new data and correcting its mistakes.
<u>F</u>	Learning is a one-time event that stops once the initial model is built.
<u>T</u>	A well-trained model can generalize its learning to identify patterns it has never seen before.
<u>T</u>	Repetition helps reinforce associations and improve pattern recognition.

Activity 1.2

Answer the questions below.

1. What is the best definition of learning?
 - a) The ability to remember everything perfectly
 - b) The ability to perform tasks without error
 - c) **The ability to form and use new associations**
 - d) The process of storing all inputs in memory

2. Which of the following is NOT a fundamental attribute of machine learning?
 - a) Recognizing patterns
 - b) Repeating examples
 - c) **Memorizing facts**
 - d) Creating associations

3. What does pattern recognition allow a system to do?
 - a) Respond only to labeled data
 - b) **Detect recurring regularities in inputs**
 - c) Ignore all differences in data
 - d) Randomly sort unrelated data

4. In pattern recognition, which kind of feature is most likely to be ignore?
 - a) The shared structure across examples
 - b) A recurring sound pattern
 - c) The repeated shape of an object
 - d) **An irrelevant design variation**

5. In supervised learning, what is given to the AI system to help it learn?
 - a) Data with missing pieces
 - b) **Data labeled with correct answers**
 - c) Completely random patterns
 - d) Unlabeled examples from multiple sources

 Activity 1.2 (continued)

Answer the questions below.

6. Which statement describes unsupervised learning?
 - a) **The AI finds patterns without labels**
 - b) The AI is told which output is correct
 - c) The AI memorizes all data examples
 - d) The AI ignores unknown patterns

7. What is a model in machine learning?
 - a) A database of all stored inputs
 - b) A system of memorized examples
 - c) A set of outputs from one training session
 - d) **An internal system of learned patterns and associations**

8. What does it mean when a model can generalize?
 - a) It copies exact answers from training data
 - b) It makes up patterns without learning
 - c) It erases data to free memory space
 - d) **It applies learned patterns to new, unseen data**

9. What happens when a model makes a misclassification?
 - a) It automatically deletes the data
 - b) The model is tested on easier data
 - c) **It is retrained or refined to improve**
 - d) The model shuts down for safety

10. Which statement best reflects the idea of continuous learning?
 - a) A model is fixed and never changes after being used
 - b) **Learning continues as the model encounters new data and errors**
 - c) Once trained, a model no longer needs updates
 - d) Learning stops once the system is deployed

Activity 1.3

Read the narrative and answer the questions below.

Plant Learning

A team of scientists is using machine learning to build a system that can identify different types of plants. At first, they collect thousands of plant images and label each one with its name (e.g., “rose,” “sunflower,” “tulip”). They use these labeled examples to teach the AI system what each plant looks like. Over time, the AI begins to recognize important patterns in the shape of the leaves and petals, while ignoring variations like background color or lighting.

Once the model is built, it is tested on a new set of plant images it has never seen before. It correctly identifies many of them but fails to recognize some roses that were photographed from an unusual angle. The scientists examine these mistakes and add more examples showing roses from different angles. The system updates its model and improves its accuracy.

1. What type of learning is being used to teach the AI system to identify plants, and how do you know?

Supervised learning, because the AI system was trained on labeled data (each image had the plant's name).

2. What features did the model learn to focus on, and which did it learn to ignore?

It learned to focus on shapes of leaves and petals (important features) and to ignore background color and lighting (irrelevant features).

3. Why is it important to have the model tested on images it had never seen before?

To check if it could generalize — apply what it learned to new situations.

4. Why did the model fail to recognize some of the roses? What was missing?

The model failed because it hadn't seen enough variety in angle or perspective during training, so it didn't recognize roses in unfamiliar views.

5. How did the model improve, and what process does this describe?

The model improved through training/refinement — it learned from its mistakes and updated its internal representation to become more accurate.

Activity 1.4

Read the narrative and answer the questions below.

Calls of the Wild

Scientists are studying animal sounds recorded in a rainforest. They have thousands of short audio clips, but most of them are not labeled — they don't know which species made each sound. Instead of guessing or labeling by hand, they use unsupervised learning to help them organize the data.

The AI analyzes the audio clips and looks for recurring patterns in pitch, rhythm, and frequency. Over time, it groups sounds that are similar into distinct clusters. Each cluster represents a sound pattern group, but the computer does not know what animal made which sound.

By grouping the data this way, the AI helps scientists discover structure in the sound recordings. This allows researchers to focus their time on analyzing each group and potentially identifying unknown species or behaviors.

1. What kind of learning is the AI using in this scenario, and why?

Unsupervised learning, because the AI is working with unlabeled sound data and grouping it based on similarity.

2. What features of the sounds is the AI analyzing to group them?

The AI analyzes features like pitch, rhythm, and frequency of the sounds.

3. Why doesn't the AI label the sounds during the learning process?

Because it was not given labels — it doesn't know which species made which sound.

4. What does the AI produce at the end — labels or something else?

The AI produces clusters or groups of similar sounds, not labeled categories.

5. How does this help scientists who are studying animals in the wild?

It helps scientists by organizing large amounts of unlabeled data so they can focus on patterns, identify unknown species, or find new behaviors.



Activity 1.5

In this activity, you'll explore how to build, test, and use a machine learning model using an online platform. You'll start by uploading sample images provided to you and then have the option to add your own examples.

Choose a Platform

Start by selecting a simple machine learning platform. Two recommended options are:

- Teachable Machine by Google
- Machine Learning for Kids

Both platforms allow you to upload data, train a model, and test it — but Machine Learning for Kids also supports model refinement and feedback.

Getting Started

- Choose a platform.
- Create a new image project.
- Upload the provided images to build at least two classes (e.g., “cat” and “dog”).
- Click Train Model.
- Test your model using new images.
- (Optional) If the model makes mistakes, add more examples and retrain.

Sample Image Sets

To help you get started, choose one of the following sets. Each contains enough examples to form two distinct classes:

- Cats and Dogs: bit.ly/cats_dogs_ML
- Flowers and Trees: bit.ly/flowers_trees_ML

You may also create your own dataset by taking or drawing pictures and uploading them to the platform.

LESSON 2: PREDICTION

In the previous lesson, we explored how an AI system learns using machine learning. It analyzes data, identifies patterns, and builds a model that helps it understand and make sense of new inputs. For example, after studying many examples, the AI forms an internal model of what a 'cup' looks like. This model becomes the basis for making predictions when it encounters a new object.

In this lesson, we will explore the basics of prediction and how Big Data helps AI make more accurate and reliable predictions.

PREDICTION

The answer to something is not always obvious. In many situations, the outcome isn't clear, and we have to make a guess. When we do, we rely on our experience and knowledge to make the best guess we can — even if we're not completely sure.

Prediction is the process by which an intelligent system — including humans — tries to estimate the most likely answer or outcome in a new situation.

PREDICTION IS WHEN AN AI USES ITS MODEL TO ESTIMATE THE MOST PROBABLE OUTPUT.

EXAMPLE 2.1

When Maya first learned to associate the object <cup> with the word 'cup', her mental model was very simple — it was based on just one cup. At first, she guessed correctly when she saw similar-looking cups. But when she saw cups that looked different — maybe taller, wider, or with a lid — she sometimes guessed wrong.

Each time she made a wrong prediction, her parents corrected her. Maya learned from her mistake and updated her model of what a cup could look like.

Over time, with more examples and feedback, her predictions improved. She became better at correctly classifying new objects as cup or not a cup — even if they looked different from the first one she saw.



Maya is learning to recognize cups. With each new example, her ability to recognize cups improves — increasing the probability that she will predict correctly, even when they look different.

In practice, prediction in AI takes different forms depending on the type of data it works with. In general, prediction applies to two kinds of data: categorical and continuous. In both cases, the AI draws on patterns it has learned from previous data to make its best guess — a prediction that reflects the most probable outcome given the information it has.

Whether it is predicting a category or estimating a numerical value, the model is performing the same essential task: using what it has learned to make a well-informed prediction in a new situation.

CATEGORICAL PREDICTION

Sometimes, the goal is to decide which category something belongs to. For example, the AI might try to recognize whether an object is a cat, determine the language of a spoken word, or match an animal sound to its species. These are all examples of categorical prediction, where the AI chooses the most likely label from a set of possible categories — much like answering the question, “What is this?”

EXAMPLE 2.2

At a recycling facility, a robotic system separates plastic bottles from other waste. A camera captures images of objects on a moving conveyor belt and sends them to a prediction model trained to recognize plastic bottles. The model analyzes each image and estimates the probability that the object is a plastic bottle. If the probability is high enough, the system triggers a robotic arm to remove the object and place it in the plastic recycling bin.



An AI-controlled robotic arm identifies and removes a plastic bottle from a moving conveyor belt at a recycling facility.

When the system began operating, it was tested with 100 items — 50 plastic bottles and 50 non-bottle items. During the test, the AI model made 30 mistakes.

- It failed to identify 20 plastic bottles, meaning it correctly classified only 30 out of 50 bottles.
- It mistakenly classified 10 non-bottle items as plastic bottles, meaning it correctly identified 40 non-bottle items as not bottles.

In total, the system got 70 out of 100 predictions correct, resulting in an accuracy of 70%.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} = \frac{30 + 40}{100} = \frac{70}{100} = 70\%$$

The items that were misclassified were collected and reviewed to identify the reasons for these errors, helping the team understand how the model could be improved.

- **Bottles Missed:** The model failed to recognize green bottles and crumpled bottles, which looked different from the training examples.
- **Objects Misclassified:** The model incorrectly classified these objects as plastic bottles due to similar shapes or features.



Original Data

Bottles it was trained on.



Bottles Missed

The model missed green bottles and crumpled bottles.



Objects Misclassified

The model misclassified objects that had similar shapes to bottles it was trained on.

ACCURACY IS ONE MEASURE OF A MODEL'S PERFORMANCE.

To improve the model's predictive accuracy and reduce its errors, the AI was retrained using a larger and more diverse set of images. This new training data included:

- Bottles in various colors (green, blue, and clear)
- Bottles that were crushed, dented, or partially obscured
- Images taken from different angles and lighting conditions
- More examples of non-bottle items with similar shapes or materials

By increasing the volume and variety of training data, the model learned to recognize a wider range of bottles and distinguish them more reliably from other objects.

The updated model was tested again using a new set of 100 items, with the same distribution as before: 50 plastic bottles and 50 non-bottles. This time, the model made only 10 mistakes:

- It missed 5 actual plastic bottles,
- And it misclassified 5 non-bottles as bottles.

That means it correctly identified 45 plastic bottles and 45 non-bottle items, giving a new accuracy of:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} = \frac{45 + 45}{100} = \frac{90}{100} = 90\%$$

GREATER VOLUME AND VARIETY OF DATA IMPROVE A MODEL'S PREDICTIVE ACCURACY.

CONTINUOUS PREDICTION

In the previous section, we explored categorical prediction, which applies to categorical data and involves assigning inputs to predefined categories. But in many cases, the task is not to classify, but to estimate a number. The AI might be asked to predict tomorrow's temperature, estimate how many points a student will score on a test, or forecast how much it will rain.

These are all examples of continuous prediction, where the goal is not to choose a category but to estimate a numerical value — answering questions like “How much?” or “What value?”

EXAMPLE 2.3

In a fictional neighborhood, an AI system is being developed to help estimate house prices based on their size. The only information the AI receives is the floor area of each house (in square meters) and the price it sold for. The goal is to train the AI to recognize the pattern between size and price so that it can predict the price of a new house based only on its size.

To begin, the AI is trained on a small dataset containing the sizes and sale prices of just six houses. Using this limited information, the AI attempts to learn the relationship between a house's size and its price by identifying a pattern in the data.



Size (square meters)	Price (\$1,000's)
107	291
115	292
107	302
115	305
93	227
82	173
100	274
120	340
130	316
90	234

This figure shows the data used to teach the AI model how house size relates to price. The scatter plot displays the actual data points, while the red dashed line represents the best-fit line created by the model. The table lists the size and sale price of the ten houses used to train the model.

The model derives the equation for the best-fit line and uses it to predict prices based on house size. This equation allows the AI to estimate the selling price of any house in the neighborhood, based solely on its size. In this case, the equation is:

$$\text{Price} = 2.53 \times \text{Size} + 18.45$$

Now the model must be tested. It is given the sizes of five houses it has not encountered before. Using its best-fit line equation, the model predicts the price of each house based solely on its size.

To determine how well the prediction model performs, the predicted prices are compared to the actual sales prices. The difference between each predicted and actual value is called the error.

$$\text{Error} = \text{Actual Price} - \text{Predicted Price}$$

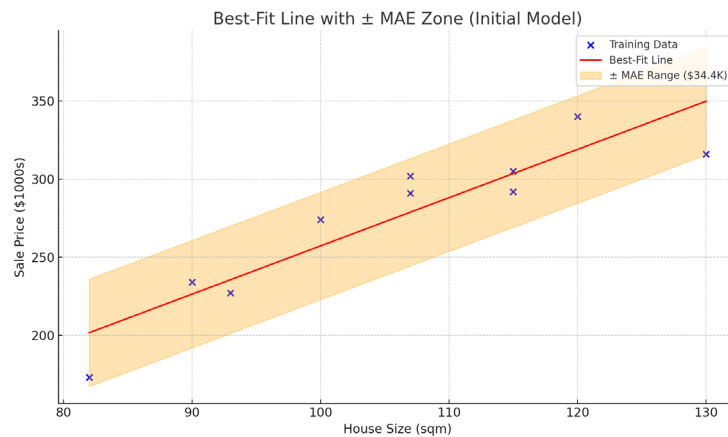
Size (sqm)	Actual Price (\$1,000)	Predicted Price (\$1,000)	Error (\$1,000)
111	291	124	-52
124	331	316	-15
157	433	406	-27
114	300	264	-36
146	399	357	-42

This table shows the actual and predicted prices for five houses that were not part of the training data. The error is calculated by subtracting the predicted price from the actual price. All values are negative, indicating that the model consistently underestimated the true prices.

A positive error means the model underestimated the price, while a negative error means it overestimated. To measure overall performance, we calculate the Mean Absolute Error (MAE) — the average size of the errors, using their absolute values to ignore direction. This tells us how far off the predictions are, on average.

$$\text{MAE} = \frac{\text{Sum of absolute value of errors}}{\text{Total Predictions}} = \frac{52 + 15 + 27 + 36 + 42}{5} = 34.4$$

The best-fit line shows what the model predicts, but the real-world value won't always land exactly on that line. The MAE gives us a typical margin of error, so we expect actual prices to fall within a band around the line — about plus or minus the MAE.



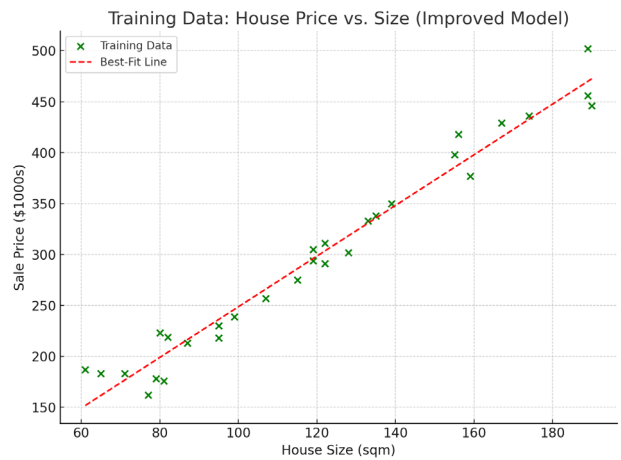
This graph shows the best-fit line produced by the model. The orange shaded area represents the full Mean Absolute Error (MAE) of \$34,400, illustrating the average range within which actual prices may fall around each predicted value.

THE MEAN ABSOLUTE ERROR IS ONE MEASURE OF A MODEL'S PERFORMANCE.

To improve the predictive performance of the model, it is trained on a larger volume and more varied set of data. Not only was it trained on a greater number of houses, but also on a wider range of sizes and prices. In the initial dataset, house sizes ranged from 82 to 130 square meters. Now, the model is exposed to houses both smaller and larger, with sizes ranging from 61 to 190 square meters.

The model recalculates the best-fit line equation using the expanded dataset and applies it to generate new predictions.

$$\text{Price} = 2.49 \times \text{Size} + 0.14$$



This graph shows the expanded training dataset, consisting of 30 houses with sizes ranging from 61 to 190 square meters. Each green point represents one house, and the red dashed line is the best-fit line generated by the model to represent the relationship between house size and price.

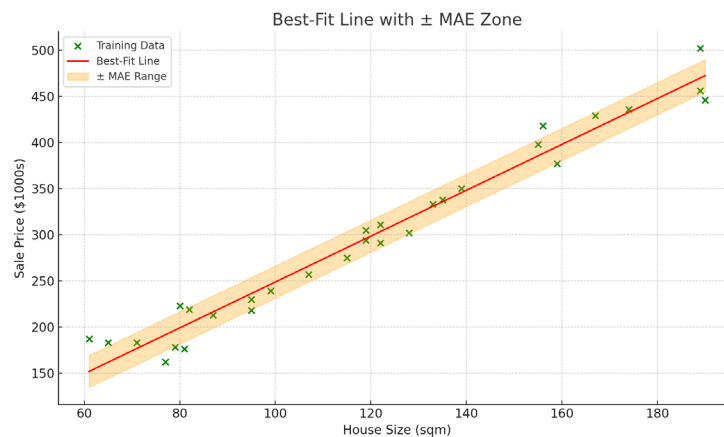
Using the new best-fit line, the model is tested again on the same five house sizes as before. For each, it predicts a price, which is then compared to the actual sale price. This allows us to calculate a new MAE, and evaluate whether the model’s performance has improved.

Size (sqm)	Actual Price (\$1,000)	Predicted Price (\$1,000)	Error (\$1,000)
111	239	276	-37
124	316	308	8
157	406	390	16
114	264	284	-20
146	357	363	-6

This table shows the model’s predictions after retraining on a larger dataset. For each house, the predicted price is compared to the actual sale price, and the error is calculated. Positive errors indicate underestimation; negative errors indicate overestimation.

$$MAE = \frac{\text{Sum of absolute value of errors}}{\text{Total Predictions}} = \frac{37 + 8 + 16 + 20 + 6}{5} = 17.4$$

As the volume and variety of training data increase, the MAE decreases. This means the zone of error around the best-fit line becomes narrower, making the model's predictions closer to the actual values.



This graph shows the best-fit line generated by the improved model, trained on a larger and more varied dataset. The orange shaded area represents the full Mean Absolute Error (MAE) of \$17,400 — a much narrower range than in the previous model, which had an MAE of \$34,400. The reduced error band indicates that the model's predictions are now, on average, closer to the actual sale prices.

Notes

Common Misconceptions

- An error means a bad prediction: All models make errors. The goal is to minimize them, not eliminate them — errors help the model improve.
- More data means a perfect model: More data helps, but only when it includes useful variety. Repetition or poor-quality data can make things worse.
- Models only learn from correct predictions: Models learn from all outcomes — including incorrect predictions — by using errors to adjust and improve over time.

CONCLUSION

In this lesson, we explored how AI makes predictions using two types of data: categorical and continuous. For categorical data, we measure how well the model performs using accuracy — the percentage of correct predictions. For continuous data, we use Mean Absolute Error (MAE) to calculate how far off the predictions are, on average.

We also saw that an AI model's ability to make accurate predictions depends on the data it learns from. As we increased the volume of training data and introduced greater variety, the model's predictions became more reliable. These two ideas — volume and variety — are part of what we call Big Data, which plays a key role in improving AI performance.

SUMMARY

Prediction is the process of estimating an outcome based on patterns learned from data.

AI can make predictions using:

- **Categorical** data (e.g., identifying whether an object is a bottle)
- **Continuous** data (e.g., estimating house prices or rainfall)

Accuracy is used to evaluate predictions on categorical data by measuring how many were correct out of all attempts.

Mean Absolute Error (MAE) is used for continuous data to measure how far off the predictions are, on average.

All models make errors — the goal is to reduce them, not eliminate them.

AI models improve as they are trained on more data. Specifically:

- **Volume** (more data) helps the model learn broader patterns.
- **Variety** (more types of examples) helps the model generalize better.

These concepts are part of what is called **Big Data**, which supports better learning and more accurate predictions in AI systems.

Activity 2.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	AI models can make predictions using either categorical or continuous data.
<u>F</u>	Accuracy is the best way to measure how close a numerical prediction is to the actual value.
<u>T</u>	Mean Absolute Error (MAE) tells us the average distance between predicted and actual values.
<u>F</u>	A model that makes prediction errors is considered broken.
<u>F</u>	More data always makes a model perfect.
<u>T</u>	Volume refers to the total amount of training data available to the model.
<u>F</u>	Variety means training a model on examples that are all the same type.
<u>T</u>	A prediction model for continuous data will produce a number, not a label.
<u>F</u>	If a model has high accuracy, that means its MAE is also low.
<u>T</u>	Big Data helps improve AI performance by increasing both volume and variety of training data.

Activity 2.2

Answer the questions below.

1. What is the goal of prediction in AI?
 - a) To memorize all data exactly
 - b) To estimate the most likely outcome using learned patterns
 - c) To randomly guess based on probability
 - d) To classify only images
2. Which of the following is an example of a categorical prediction?
 - a) Estimating the temperature tomorrow
 - b) Predicting a student's exam score
 - c) Determining if an object is a plastic bottle
 - d) Calculating total sales for a store
3. Which metric can be used to evaluate continuous predictions like temperature?
 - a) Accuracy
 - b) Precision
 - c) Recall
 - d) Mean Absolute Error (MAE)
4. What does a high MAE value tell us about a model's predictions?
 - a) The model never makes mistakes
 - b) The model's predictions are far from actual values
 - c) The model's predictions are always correct
 - d) The model is overfitting
5. How is MAE calculated?
 - a) Percentage of correct predictions
 - b) Average of the absolute differences between predicted and actual values
 - c) Number of categories predicted correctly
 - d) The square root of total errors

Activity 2.2 (continued)

Answer the questions below.

6. What does the term "volume" refer to in the context of training data?
 - a) **The total amount of data the model learns from**
 - b) The range of model outputs
 - c) The number of times a model makes errors
 - d) The speed at which the data is processed

7. Why is "variety" in training data important for improving a model?
 - a) It increases the speed of training
 - b) It reduces the cost of data collection
 - c) It prevents the use of continuous prediction
 - d) **It helps the model generalize to different examples**

8. Which best describes the difference between accuracy and MAE?
 - a) MAE is used for text predictions; accuracy is used for numbers
 - b) Both are used for categorical predictions
 - c) **Accuracy is used for categorical data; MAE is used for continuous data**
 - d) Accuracy and MAE mean the same thing

9. What happens when a model is trained on a small, narrow dataset?
 - a) It performs well on all tasks
 - b) It automatically adjusts its error
 - c) It always predicts perfectly
 - d) **It struggles to make accurate predictions on new data**

10. Which of the following statements is true?
 - a) Models only learn from correct predictions
 - b) **Models learn from both correct and incorrect predictions**
 - c) Models stop learning after one training cycle
 - d) Models only learn from testing data

Activity 2.3

Read the narrative and answer the questions below.

A software company is developing an AI system that can automatically detect the language of short text messages. To train the model, engineers collect thousands of short sentences labeled with the language they are written in — such as English, Spanish, Arabic, and French.

The model learns to recognize specific features like vocabulary, accents, and sentence structure. Once trained, it is tested on new sentences submitted by users. It correctly identifies most languages, but it struggles with shorter messages or those that use informal spelling.

After reviewing the errors, the engineers realize that the AI hadn't been trained on enough casual or abbreviated text. They improve the model by adding more examples that include informal language, slang, and abbreviations. The system becomes more accurate and consistent in real-world use.

1. What type of learning was used to train the AI system to recognize languages, and how do you know?

Supervised learning — because the model was trained using labeled examples, where each sentence was tagged with its language.

2. What kind of prediction is this AI system performing?

Categorical prediction — the model is classifying each sentence into one of several languages.

3. Why did the AI model struggle with some user-submitted sentences?

Because the model hadn't been trained on enough informal or abbreviated text, so it couldn't recognize the language accurately in those cases.

4. How did the engineers improve the model?

They retrained it using more varied examples, including informal language and slang.

5. Why is it important to train the model with a wide variety of examples?

So the model can generalize and make accurate predictions, even when input appears in unfamiliar or casual forms.

Activity 2.4

Read the narrative and answer the questions below.

A transportation company is building an AI system that can estimate how long a trip will take by car. To train the model, engineers collect thousands of examples showing how long it took to travel different distances at different times of day. Each example includes the distance traveled and the actual travel time in minutes.

The AI is first trained on a small set of data — mostly from trips taken during off-peak hours. From this, it learns a basic pattern: longer distances usually mean longer travel times. But when tested on trips during rush hour, the model often predicts travel times that are too short, because it hasn't learned how traffic affects the results.

To fix this, the engineers expand the training data to include trips taken at different times of day, including busy periods with heavy traffic. The model adjusts its understanding and begins to make more accurate predictions.

1. What kind of prediction is the AI model making in this scenario?

Continuous prediction — it's estimating a numerical value (travel time in minutes).

2. Why were the model's early predictions inaccurate during rush hour?

Because the model had not been trained on examples that included traffic delays.

3. How did the engineers improve the model's performance?

They added more varied training data from different times of day.

4. What is one way to measure how far off the model's predictions are?

By calculating the Mean Absolute Error (MAE), which gives the average difference between predicted and actual travel times.

5. Why is it important to include both volume and variety in training data?

So the model can make accurate predictions across different conditions, such as heavy or light traffic.

Activity 2.5

Read the scenario below. Then answer the questions that follow. Use the provided spreadsheet. You will practice sorting, filtering, and using formulas—just like real data scientists do when they test AI models. Use the short link to access the spreadsheet: bit.ly/C4_AI_Activity2_5

- a. A grocery distribution center is testing an AI model that sorts fruit. The model decides if each item is an Apple or Not Apple. The center tests the model on a batch of 50 fruits. For each item, the AI makes a prediction. The spreadsheet shows both the actual label and the AI prediction.

1. How many predictions were correct?	36
2. How many predictions were incorrect?	14
3. What was the model's overall accuracy?	72%
4. How many apples were missed?	7
5. How many tomatoes were misclassified?	11

6. Which fruit did the AI confuse with apples? Why do you think this happened?

Tomatoes — confused with apples because they are round, similar in size, and often red like apples

- b. The grocery distribution center is considering using the AI model for full operations. On average, the center handles 100,000 apples every week, along with other fruits. When the AI makes a mistake, it costs the business money:

- Missing an apple = \$5 lost
- Mistaking a tomato for an apple = \$1 wasted

1. How many apples would be missed every week?	14,000
2. How many tomatoes would be misclassified every week?	22,000
3. What would the total weekly cost of missed apples be?	\$70,000
4. What would the total weekly cost of misclassified tomatoes be?	\$22,000
5. What is the combined weekly cost of using this AI model?	\$92,000

6. Based on these numbers, do you think the AI is ready for real-world use? Why?

The AI is not ready for operations — the cost of errors is extremely high. The model would need more training or a different approach before being deployed.

Activity 2.5 (continued)

Read the scenario below. Then answer the questions that follow. Use the provided spreadsheet. You will practice sorting, filtering, and using formulas—just like real data scientists do when they test AI models. Use the short link to access the spreadsheet: bit.ly/C4_AI_Activity2_5

- c. After reviewing the high cost of errors, the distribution center retrained its AI model to better distinguish between apples and tomatoes.

Refer to the sheet 'Activity 2.5c' to answer the following questions.

1. What is the accuracy of the new model?	90%
2. How many apples would it miss out of 100,000?	4,000
3. How many tomatoes would it misclassify out of 100,000?	6,000
4. What would the weekly cost of missed apples be?	\$20,000
5. What would the total weekly cost of misclassified tomatoes be?	\$6,000
6. What is the total weekly cost of errors with the new model?	\$26,000

7. How does this compare to the old model in Activity 2.5b? What is the improvement?

Old model cost = \$92,000 per week. New model cost = \$26,000 per week.

Savings = \$66,000 per week.

8. Think about another area of life. Where would AI errors be more serious than in a grocery store?

AI errors would be more serious in areas where lives are at stake. Examples:

Healthcare: A wrong diagnosis could harm a patient.

Traffic/Transportation: A self-driving car could cause an accident.

Airlines: An AI error in navigation or maintenance could risk hundreds of lives.

9. Imagine you are in charge of the distribution center. Would you use the 90% accurate AI, or keep humans sorting apples, or a combination of both AI and human? Explain your choice.

I would choose a combination of AI and human workers. The AI is fast and saves money, but at 90% accuracy it still makes mistakes. By adding human quality control, people can catch the errors the AI misses. In the real world, this is called 'Human-in-the-Loop AI'. AI and humans working together are more effective than either on their own.

Activity 2.6

Read the scenario below, then answer the questions that follow. Use the provided spreadsheet to apply formulas and calculations—just like real data scientists do when they test AI models. Use the short link to access the spreadsheet: bit.ly/C4_AI_Activity2_6

A company is testing an AI model to predict how many units of a product it will sell at different prices. Accurate predictions help the company plan inventory and revenue.

The AI studied data and built the following best-fit line:

$$\text{Sales (units)} = -12 \times \text{Price (\$)} + 220$$

The company then tested the model at different prices between \$8.00 and \$12.50.

- Complete the table by using the equation of the best-fit line to predict the number of units sold and the error for each prediction.

Price (\$)	Actual Units Sold (units)	Predicted Units Sold (units)	Error (units)	Error (Absolute value)
8.0	115	124	-9	9
9.0	125	112	13	13
10.0	95	100	-5	5
11.0	101	88	13	13
12.0	67	76	-9	9

- Calculate the Mean Absolute Error.

The MAE = 9.80

- If the AI makes one prediction every day, then over a 30-day month, by how many units will the company misestimate sales?

Total Error (units) = MAE x Days = 9.80 x 30 = 294 units

- If the company sells 10,000 units in a month, what percentage of sales does the total error (294 units) represent? What does this tell us about the model's usefulness?

294/10,000 x 100 = 2.94%

The model's error is about 3% of total sales. It is fairly accurate, but lowering the error could still save hundreds of units each month.

LESSON 3: LOGIC & REASONING

So far, we've seen that AI can learn from data and predict outcomes. But there is more to intelligence — reasoning. We use reasoning all the time. Much of our everyday thinking is reasoning. It helps us understand what we observe, solve problems, and decide what to do.

Reasoning, though we may not always notice it, is structured. There are rules to thinking that guide us toward conclusions. These conclusions, or outcomes of reasoning, may be an understanding, a new insight, a decision, or a plan. The rules that make reasoning work are referred to as logic.

In this lesson, we will explore reasoning, its forms, and the framework in which it operates: logic.

REASONING

Reasoning is a structured way of thinking — or processing — that helps us uncover new information, make decisions, and make sense of our world. In every case, reasoning follows a simple path: input, processing, and output. It begins with what is already known — information or knowledge that serves as the input. This is then processed, and the outcome is produced. In general, three outcomes are possible: an inference, a decision, or an explanation.

REASONING IS THE STRUCTURED WAY WE THINK TO DISCOVER,
DECIDE, OR UNDERSTAND.



INFERENCE

Inference means arriving at a new piece of information. It is the process of linking information we already have in order to reach a conclusion. The information we start with forms the basis or starting point of the reasoning process: these are called premises.

Premises may come from three sources:

1. facts we already know, which form general rules.
2. observations our senses are picking up.
3. predictions based on learned patterns.

A PREMISE IS THE STARTING POINT OF INFERENCE AND CAN BE A FACT, OBSERVATION, OR PREDICTION.

EXAMPLE 3.1 (HI - HUMAN INTELLIGENCE)

Layla is in science class, listening carefully as her teacher explains that all birds have feathers. A moment later, the teacher holds up a picture of a chicken and asks the class, “Is this a bird?” Layla thinks it through. She recalls what she has just learned: all birds have feathers. She observes that chickens have feathers. From this, she reasons that chickens must be birds.

LOGICAL FORM

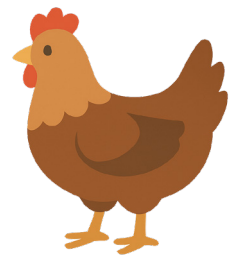
PREMISES

GENERAL RULE: ALL BIRDS HAVE FEATHERS.

OBSERVATION: CHICKENS HAVE FEATHERS.

INFERENCE

THEREFORE : CHICKENS ARE BIRDS.



INFERENCE IS THE PROCESS OF ARRIVING AT NEW INFORMATION.

EXAMPLE 3.1 (HI - ARTIFICIAL INTELLIGENCE)

A video app observes that a student has recently watched several basketball highlights. The AI model behind the app has learned a general rule from past data: *viewers who watch videos about a topic will probably enjoy other videos on that same topic.*

Using this knowledge, the system reasons that the student is likely interested in basketball more broadly. From this, it infers that the student would also enjoy basketball training videos and recommends them.



LOGICAL FORM

PREMISES

GENERAL RULE : VIEWERS WHO WATCH VIDEOS ABOUT A TOPIC WILL PROBABLY ENJOY OTHER VIDEOS ON THAT SAME TOPIC.

OBSERVATION : THE STUDENT WATCHED BASKETBALL HIGHLIGHTS.

INFERENCE

THEREFORE : PROBABLY, THE STUDENT WOULD ENJOY BASKETBALL TRAINING VIDEOS.

Both examples show how inference arrives at a new conclusion by linking what we already know. In the first case, the conclusion “*Chickens are birds*” followed from a general rule (*All birds have feathers*) combined with a specific observation (*Chickens have feathers*). Because this reasoning was grounded in a fact, the inference carried a high degree of certainty.

In the second case, the conclusion “*The student would probably enjoy basketball training videos*” followed from a general rule learned from data (*Viewers who watch videos on a topic will probably enjoy more videos on that topic*) and an observation (*The student watched basketball highlights*). Here the inference relied on a prediction, and since predictions come with uncertainty, the conclusion is expressed only as probable.

THE CERTAINTY OF AN INFERENCE DEPENDS ON THE CERTAINTY OF ITS PREMISES.

The certainty of an inference depends on the certainty of its premises. If the premises are facts, the conclusion is highly certain. If the premises are predictions, the conclusion is less certain and best understood as probably true rather than definitely true.

DECISION-MAKING

Decision-making means selecting the best choice from several possibilities. It is the process of looking at different choices, considering what will probably happen if we take each one, and then selecting the option with the highest utility — the one that is most useful, valuable, or best fits our goal.

DECISION-MAKING IS THE PROCESS OF SELECTING THE CHOICE WITH THE HIGHEST UTILITY.

Decisions are also guided by premises: facts, rules, observations, predictions, and even personal preferences or opinions. Which choice seems best depends on the context — sometimes facts dominate, other times preferences or predictions play the larger role.

EXAMPLE 3.2 (HI - HUMAN INTELLIGENCE)

Lina was getting ready for school and saw that the sky was cloudy. She thought about walking, riding her bike, or asking her father for a ride. She remembered that she didn't mind a little rain and that she really disliked depending on her father's schedule since he sometimes left very early. After weighing her options, she decided to take her bike.



LOGICAL FORM

PREMISES

PREFERENCE: I DON'T MIND A LITTLE RAIN.

PREDICTION: IT IS LIKELY THAT MY FATHER WILL LEAVE VERY EARLY.

CHOICES

CHOICE 1: IF I WALK, THEN PROBABLY I WILL BE SLOW AND MIGHT GET WET.

CHOICE 2: IF I BIKE, THEN PROBABLY I WILL ARRIVE FASTER AND RISK ONLY A LITTLE RAIN.

CHOICE 3: IF I RIDE WITH MY FATHER, THEN PROBABLY I WILL STAY DRY BUT ARRIVE TOO EARLY AT SCHOOL.

DECISION

THEREFORE: I WILL RIDE MY BIKE TO SCHOOL.

In Lina's case, her premises included a personal preference — *she didn't mind a little rain* — and a prediction — *her father will likely leave very early*. These premises shaped how she evaluated her choices.

She compared three choices: *walking*, *biking*, or *riding with her father*. Each choice came with possible outcomes, and she weighed the utility (how useful or valuable each outcome was to her).

1. Walking would take too long and might make her wet (low utility).

2. Riding with her father would keep her dry but could get her to school too early (low utility).
3. Biking was fast, comfortable enough, and matched her preference about rain (highest utility).

Her decision to ride her bike was based on comparing the outcomes against her premises and selecting the choice with the best overall value.

EXAMPLE 3.2 (AI - ARTIFICIAL INTELLIGENCE)

A self-driving car approaches an intersection. Its AI system considers three options: keep driving, slow down, or stop. Its sensors show that a pedestrian is stepping onto the crosswalk and a car is close behind. The car's decision-making is guided by two fundamental rules: never harm humans, and avoid collisions unless it contradicts the first rule. Based on this, the car decides to stop for safety.

LOGICAL FORM

PREMISES

RULE 1: NEVER HARM HUMANS.

RULE 2: AVOID COLLISIONS UNLESS IT CONTRADICTS RULE 1.

CHOICES

CHOICE 1: IF THE CAR KEEPS DRIVING, THEN THE PEDESTRIAN WILL BE HARMED.

CHOICE 2: IF THE CAR SLOWS DOWN, THEN PROBABLY IT WILL AVOID HARM BUT MAY RISK A REAR COLLISION.

CHOICE 3: IF THE CAR STOPS, THEN IT WILL NOT HARM THE PEDESTRIAN, BUT A REAR COLLISION WILL OCCUR.

DECISION

THEREFORE: STOP THE CAR.



For the self-driving car, its premises were programmed as rules: *never harm humans* and *avoid collisions unless that contradicts the first rule*. With these rules in place, the car evaluated three choices: *keep driving*, *slow down*, or *stop*. Each choice had different outcomes.

1. Continuing to drive would harm a pedestrian (very low utility).
2. Slowing down might possibly avoid harm but still risk a probable rear collision (medium utility).
3. Stopping would certainly keep the pedestrian safe even though it might cause a rear collision (highest utility, because protecting humans is the top rule).

The AI system was deciding under a “Definitely Do Not” condition set by its first rule. This condition gave that rule the highest utility or value, so it overruled the other possibilities. This shows how the car compared the possible outcomes against its guiding premises, then selected the option with the greatest utility — the one most consistent with its goals and priorities.

UTILITY IS A MEASURE OF HOW USEFUL, VALUABLE, OR GOAL-FITTING AN OUTCOME IS.

Although Lina and the self-driving car are very different, they both used the same reasoning pattern. Each began with premises — Lina’s were based on her preferences and a prediction, while the car’s were based on safety rules.

They considered several choices, weighed the possible outcomes, and compared the utility of each option. Lina’s decision was shaped by comfort, while the car’s was shaped by safety, but the structure of their reasoning was the same.

THE UTILITY OF A CHOICE DEPENDS ON THE PREMISES USED TO JUDGE IT.

EXPLANATION

Explanation is the process of reasoning backward from a given — something already accepted as true — in order to uncover its causes, contributing conditions, or relationships. In other words, explanation begins with an outcome and asks: “What conditions could have produced this?” Instead of starting with premises and moving forward, as in inference, explanation works in reverse — from an observation to possible reasons. In terms of directionality, explanation is therefore the opposite of inference.

A GIVEN IS THE STARTING POINT OF EXPLANATION, FROM WHICH WE REASON BACKWARD TO UNCOVER ITS CAUSES, CONTRIBUTORS, AND RELATIONSHIPS.

In terms of strategy, however, explanation is closer to decision-making: both involve a search process. Decision-making searches among possible actions to select the one with the highest utility, while explanation searches among possible reasons to identify the one with the greatest plausibility.

THE PLAUSIBILITY OF AN EXPLANATION DEPENDS ON THE PREMISES THAT SUPPORT IT.

Like inference and decision-making, explanation depends on a knowledge base — the rules, facts, or learned patterns that connect conditions to outcomes. Without this background knowledge, reasoning backward would not be possible. But unlike inference, which can sometimes deliver definite conclusions, explanation typically carries varying degrees of uncertainty. The strength of an explanation depends on the plausibility of its premises and how well they fit the observed outcome.

EXAMPLE 3.3 (HI - HUMAN INTELLIGENCE)

Omar steps outside one morning and notices that the front lawn is wet. He wonders what caused it. Two possibilities come to mind: maybe it rained during the night, or maybe the sprinkler was on. Omar thinks about what usually happens: when it rains, everything in the neighborhood gets wet, not just his lawn. But today, he notices that only the front lawn is wet while the rest of the ground is dry. Based on this observation, Omar reasons that the most likely cause is that the sprinkler was on.

LOGICAL FORM

GIVEN

OBSERVATION: THE FRONT LAWN IS WET.

PREMISES

GENERAL RULE 1: IF IT RAINS THEN EVERYTHING IS WET.

GENERAL RULE 2: IF THE SPRINKLER IS ON THEN ONLY THE FRONT LAWN GETS WET.

OPTIONS

OPTION 1: THE CAUSE MIGHT BE RAIN.

OPTION 2: THE CAUSE MIGHT BE THE SPRINKLER.

OBSERVATION: ONLY THE FRONT LAWN IS WET.

EXPLANATION

THEREFORE: SINCE ONLY THE FRONT LAWN IS WET, THE MOST PROBABLE EXPLANATION IS THE SPRINKLER WAS ON.



This example shows how explanation works by reasoning backward from an outcome. Omar begins with what is given — the lawn is wet — and considers different possible reasons. By comparing these options against additional evidence, he chooses the cause that best fits.

EXAMPLE 3.3 (AI - ARTIFICIAL INTELLIGENCE)

A medical AI system is fed patient information: the person has both a fever and a cough. The system compares this data to patterns in its knowledge base. It “knows” that the flu typically produces both fever and cough, while allergies can produce cough but not fever. Since the patient’s data includes fever along with cough, the AI rules out allergies as the cause and identifies the flu as the most probable explanation.



LOGICAL FORM

GIVEN

OBSERVATION: THE PATIENT HAS A FEVER AND COUGH.

PREMISES

GENERAL RULE 1: IF A PATIENT HAS THE FLU, THEN FEVER AND COUGH OCCUR TOGETHER.

GENERAL RULE 2: IF A PATIENT HAS ALLERGIES, THEN COUGH OCCURS BUT FEVER DOES NOT.

OPTIONS

OPTION 1: THE CAUSE MIGHT BE FLU.

OPTION 2: THE CAUSE MIGHT BE ALLERGIES.

OBSERVATION: THE PATIENT HAS BOTH FEVER AND COUGH.

EXPLANATION

THEREFORE: SINCE THE PATIENT HAS BOTH FEVER AND COUGH, THE MOST PROBABLE EXPLANATION IS FLU.

The AI follows the same reasoning process. It starts with the observed symptoms — fever and cough — and evaluates possible reasons using its knowledge base. By ruling out what doesn’t fit, it selects the most probable explanation.

Because different conditions can lead to the same outcome, explanation is not always certain. It is often probabilistic: the best explanation is the one that fits the evidence most strongly, though other explanations may remain possible.

EXAMPLE 3.4 (HI - HUMAN INTELLIGENCE)

Omar sees his front lawn is wet. He considers two possible causes: rain or the sprinkler. Since only the front lawn is wet, he concludes the sprinkler was most likely the cause.

Later, Omar discovers that his children had been playing with water. This new possibility was not part of his original reasoning. With this added information, the sprinkler explanation loses plausibility, and the children's play becomes the best explanation.



EXPLANATIONS CAN CHANGE WHEN NEW POSSIBILITIES ARISE

EXAMPLE 3.4 (AI - ARTIFICIAL INTELLIGENCE)

A medical AI is fed information about a patient who has both fever and cough. From its knowledge base, it considers flu and allergies as possible causes. Since allergies don't usually produce fever, the AI concludes flu is the most probable explanation.

But when the system receives additional information — the patient also has a rash — the plausibility of flu decreases. The new symptom suggests another possible condition, measles, and the explanation shifts to account for this more fitting cause.



CONCLUSION

Inference, decision-making, and explanation all share common ground: they are forms of reasoning that begin with premises — facts, observations, or predictions — and apply a knowledge base to reach conclusions. Each can operate with certainty or with varying degrees of probability.

Where they differ is in direction and purpose. Inference moves directly forward from premises to outcomes, helping us discover new information. Decision-making searches among possible actions to select the one that best meets a goal. Explanation searches backward from an outcome to identify the most plausible cause.

Reasoning takes many forms — whether discovering new information, choosing the best action, or searching for the most plausible cause — it is how we make sense of the world and act within it.

Notes

Common Misconceptions

- *Reasoning outcomes are always certain if the system is well-designed.*
Even with perfect logic, uncertainty arises from incomplete knowledge, noisy data, or ambiguous premises.
- *Decision-making is just picking the highest probability outcome.*
Decision-making is about expected utility, not just probability — the best choice balances both likelihood and value of outcomes.
- *Explanations must be causal.*
In AI, many explanations are associative or model-based rather than strictly causal.
- *One explanation is sufficient.*
Explanation is a search problem: multiple explanations may exist, and plausibility depends on background knowledge and available evidence.

SUMMARY

Reasoning – The structured way we think to discover, decide, or understand.

- Forms of reasoning – inference, decision, and explanation.
- Logic – the rules that guide reasoning.
- Knowledge Base – All forms depend on a knowledge base — rules, facts, or learned patterns that connect conditions to outcomes.

Inference – Arriving at a new piece of information by linking what we already know.

- Starting Point – Premises; can be:
 - Facts – general rules already known.
 - Observations – information detected through our senses.
 - Predictions – expectations based on learned patterns.
- Outcome – New information (a conclusion).
- Strategy – Direct reasoning from premises to outcome (no search).
- Directionality – Forward reasoning.
- Certainty – The certainty of an inference depends on the certainty of its premises.
 - If premises are facts, then inference is definite.
 - If premises are predictions, then inference is probable.

Decision-making – The process of selecting the option with the highest utility (the most useful, valuable, or goal-fitting).

- Starting Point – Premises (facts, rules, observations, predictions, preferences, opinions).
- Outcome – The choice with the highest utility.
- Strategy – Search among possible choices.
- Directionality – Forward reasoning.
- Utility – A measure of how useful, valuable, or goal-fitting an outcome is. It depends on the premises used to judge it.

Explanation – The process of reasoning from a outcome to uncover its causes, or contributing conditions.

- Starting Point – A given (an accepted observation or outcome)..
- Outcome – The reason with the highest plausibility.
- Strategy – Search among possible reasons.
- Directionality – Backward reasoning.
- Utility – A measure of how reasonable or believable an explanation is. It depends on the premises used to support it.

Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u> T </u>	Reasoning is a structured way of thinking that helps us discover new information, choose actions, or make sense of outcomes.
<u> F </u>	Premises are always absolute facts; observations and predictions cannot be used as premises.
<u> T </u>	The certainty of an inference depends on the certainty of its premises.
<u> F </u>	Inference involves a search among different possibilities before reaching a conclusion.
<u> T </u>	Decision-making is the process of selecting the option with the highest utility.
<u> T </u>	Utility means the usefulness, value, or goal-fit of an outcome.
<u> T </u>	Decision-making and explanation both involve a search strategy, while inference does not.
<u> T </u>	Explanation is reasoning backward from a given outcome to uncover possible conditions or reasons.
<u> T </u>	The plausibility of an explanation depends on how well its premises fit the observed outcome.
<u> F </u>	Explanations are always certain, because once the cause is found it cannot be replaced.

Activity 3.2

Answer the questions below.

1. Reasoning can best be described as:
 - a) Guessing based on experience
 - b) Remembering facts from memory
 - c) A structured way of thinking that follows rules to reach conclusions
 - d) Choosing randomly when uncertain

2. In the reasoning process, the starting points we use are called:
 - a) Outcomes
 - b) Premises
 - c) Hypotheses
 - d) Causes

3. In reasoning, logic refers to:
 - a) The memory of facts and rules
 - b) The outcome of decision-making
 - c) The search among possible causes
 - d) The rules that guide reasoning from premises to conclusions

4. Why is logic important in reasoning?
 - a) It ensures all decisions are fast
 - b) It provides structure so conclusions follow consistently from premises
 - c) It allows us to guess without evidence
 - d) It reduces the number of premises needed

5. Which of the following can be premises in inference?
 - a) Facts
 - b) Observations
 - c) Predictions
 - d) All of the above

Activity 3.2 (continued)

Answer the questions below.

6. The certainty of an inference depends on:
 - a) **The certainty of its premises**
 - b) The number of options available
 - c) The directionality of reasoning
 - d) The search process used

7. Decision-making is defined as:
 - a) Selecting the option with the highest plausibility
 - b) Reasoning backward from an outcome to causes
 - c) Directly linking premises to conclusions
 - d) **Selecting the option with the highest utility**

8. In decision-making, utility refers to:
 - a) How probable the outcome is
 - b) **How useful, valuable, or goal-fitting an outcome is**
 - c) How fast the decision can be made
 - d) How certain the premises are

9. Explanation differs from inference mainly because:
 - a) It moves forward from premises to outcomes
 - b) It depends only on facts
 - c) It guarantees certainty
 - d) **It reasons backward from a given outcome to possible conditions**

10. The starting point of explanation is called:
 - a) Premise
 - b) **Given**
 - c) Utility
 - d) Hypothesis

 Activity 3.2 (continued)

Answer the questions below.

11. The plausibility of an explanation depends on:
 - a) The number of possible outcomes
 - b) **The premises that support it and how well they fit the observed outcome**
 - c) The search method used
 - d) Whether the explanation is causal

12. Which of the following requires a search strategy?
 - a) Inference only
 - b) **Decision-making and explanation**
 - c) Explanation only
 - d) Inference, decision-making, and explanation

13. Which of the following best shows inference?
 - a) Deciding which bus to take to school
 - b) Explaining why the lawn is wet
 - c) **Linking a rule and an observation to reach a new conclusion**
 - d) Remembering a fact from memory

14. In decision-making, the best option is chosen by:
 - a) Selecting the one supported by the most facts
 - b) **Selecting the one with the highest utility**
 - c) Selecting the one with the highest probability of occurring
 - d) Selecting the one that requires the fewest premises

15. Why is uncertainty common in explanation?
 - a) Because explanations always require predictions
 - b) **Because multiple possible reasons can fit the same outcome**
 - c) Because causes never align with outcomes
 - d) Because it does not use a knowledge base

Activity 3.3

Read the narrative and answer the questions below.

- a.** In science class, students were learning about photosynthesis. The teacher explained that all plants require sunlight to survive. She then instructed them to start an experiment by placing one plant (A) near the window and another plant (B) inside the closet.

1. Infer the outcome for Plant (A).

Plant (A) will survive.

2. Infer the outcome for Plant (B).

Plant (B) will not survive.

3. What is the general rule you used to make this inference?

All plants require sunlight to survive.

4. State your logic for inferring the outcome for Plant (B).

Since all plants require sunlight to survive, and Plant (B) will not get any sunlight, therefore Plant (B) will not survive.

- b.** An AI model was trained on thousands of cat images to recognize features of cats. From the training data, it learned the general rule that cats usually have pointed ears and whiskers. The model is then shown a new image, which it classifies as a cat.

1. What form of reasoning did the AI model use to make this prediction?

Inference

2. What general rule did it acquire from its training?

If an animal has pointed ears and whiskers, then it is probably a cat.

3. Would you consider this general rule one of certainty?

No, it is not 100% certain. Some breeds of cats do not have pointed ears, so the rule is probabilistic rather than definite.

Activity 3.4

Read the narrative and answer the questions below.

- a. Amir is deciding how to spend his evening. He has two options: study for his math test tomorrow or watch a movie and go to bed early. He knows that studying will help him do better on tomorrow's test, but he also wants to relax after a long day.

1. What form of reasoning is Amir using?

Decision-making

2. What premises is Amir considering to make a choice?

His goal to do well on the test and his preference for relaxation.

3. What outcome does decision-making seek to produce?

The choice with the highest utility.

4. If Amir chooses to study, what does this reveal about his values?

He values doing well on the test more than relaxation.

- b. A delivery drone approaches a tall building on its route. It must decide whether to fly over the building, or around it. Its rules are: (1) minimize delivery time, and (2) conserve battery when possible. The drone maps the routes and calculates that flying over the building will take less time but will require more energy, while flying around it will require more time but less energy.

1. What premises guide the drone's reasoning?

Rules about minimizing delivery time and conserving battery.

2. What other information does the drone need to make its choice?

Which rule is more important in this situation.

3. If the drone chooses to fly around the building, what does this tell you about its programmed priorities?

It assigns higher utility to conserving battery than to minimizing time.

Activity 3.5

Read the narrative and answer the questions below.

- a. Sara walks into the kitchen and sees that the cookie jar is empty. She considers two possibilities: her brother ate the cookies, or her mother gave them to the neighbors. Sara looks around for clues, she notices cookie crumbs on the table.

1. What form of reasoning is Sara using?

Explanation

2. What is the given in this scenario?

The observation that the cookie jar is empty.

3. What possible reasons does Sara consider?

Her brother ate the cookies, or her mother gave them away.

4. Which explanation is most plausible, and why?

The explanation that her brother ate them is most plausible, because the crumbs on the table support this outcome.

- b. A smart home system detects that the living room lights are on in the middle of the night. From past data, it considers two possibilities: someone is awake and moving around, or the timer system malfunctioned. It also detects motion in the living room.

1. What is the given in this scenario?

The observation that the living room lights are on in the middle of the night.

2. What possible reasons does the AI consider?

That a person is awake, or that the timer system malfunctioned.

3. Which explanation is most plausible, and why?

The explanation that a person is awake is most plausible, because the motion sensor confirms activity.

Activity 3.6 - Practicing Reasoning with AI

In this activity, you will interact with an AI which will act as a mentor, providing questions and explanations to deepen your understanding of the topics of the lesson.

In the shared folder you will find five separate text documents, each containing the prompt you need for the activity, as well as the PDF of the lesson summary.

Access the folder from: bit.ly/C4_AI_Activity3_6

a. Reasoning Scenarios

In this part, you will practice three forms of reasoning:

- Inference
- Decision-Making
- Explanation

Each reasoning type has its own text file with a prompt.

Steps to Follow:

1. Download the file from the shared Google folder and choose the prompt for the reasoning type you want to practice.
2. Copy the full prompt and paste it into ChatGPT (or another LLM).
3. The prompt will guide the AI to give you three scenarios. For each scenario, carefully read the setup.
4. Type your answer directly into the chat, always including:
 - Your answer: the outcome, action, or cause.
 - Your reason: why your answer follows from the given rule, priorities, or causes.
5. The AI will confirm if your answer is correct and evaluate whether your reasoning is sound.
6. Continue through the scenarios one at a time until you complete all three.

Tips:

Think about the structure of reasoning:

- Is it predicting? (Inference)
- Is it choosing? (Decision-making)
- Is it explaining? (Explanation)

Always explain your reasoning, even if unsure — that is what the AI will evaluate.

Don't rush; reflect on why a certain answer follows from the premises.

b. Questions & Answers

In this part, you will test your understanding of the lesson using the PDF provided in the folder. You can choose between:

- True/False Questions
- Multiple-Choice Questions

Each activity has its own text file with a prompt.

Steps to Follow:

1. Open the shared folder and download the PDF of the lesson summary.
2. Attach the PDF in ChatGPT.
3. Copy the prompt for the activity you want to practice (True/False or Multiple Choice) and paste it into ChatGPT.
4. The AI will create 10 questions based on the summary:
 - If you chose True/False, each question will be answered with “True” or “False.”.
 - If you chose Multiple Choice, each question will have 4 options (A, B, C, D).
5. For each question:
 - Read carefully and submit your answer.
 - The AI will tell you if you are correct, provide the correct answer if needed, and give a short explanation.
6. After all questions:
 - The AI will list the questions you got wrong.
 - It will give you a short review of the key topics you struggled with.
 - It will suggest a follow-up exercise to strengthen your understanding.

Tips:

- For True/False: think about why the statement is true or false before answering.
- For Multiple Choice: read all four options carefully before choosing.
- Use the AI’s feedback to focus on the concepts you find most difficult.
- Repeat the activity for more practice — the AI can generate new sets of questions each time.

LESSON 4: AI IN THE REAL-WORLD

Artificial Intelligence (AI) is no longer just a futuristic idea from science fiction. It is already part of our everyday lives, often in ways we do not even notice. From recommending what video to watch next to helping doctors detect diseases earlier, AI systems are shaping how people live, work, and learn.

In this lesson, we will explore how AI is used in the real world, what challenges and limitations it faces, and why ethics and responsibility are central to its development. By the end, you should be able to explain examples of AI applications, recognize where AI might fail, and discuss the impact AI has on individuals and society.

APPLICATIONS OF AI

AI technologies are designed to extend the very processes of intelligence we studied in earlier lessons. They can recognize patterns in data, form associations between pieces of information, and learn from past examples. With these abilities, AI systems can make predictions, classify new situations, and even carry out forms of reasoning such as inference, decision-making, and explanation. Because these are the same foundations of human intelligence, AI can be applied across many areas of daily life, industry, and science.

Most of today's systems fall under Narrow AI, which means they are designed with a narrow focus. They can perform a single task or a limited set of tasks very effectively. Examples include navigation apps that suggest the fastest route or chatbots that respond to customer service questions. Although they may appear intelligent, their abilities are restricted to their specific domain and cannot be transferred to entirely different tasks.

NARROW AI SYSTEMS ARE DESIGNED TO PERFORM ONE SPECIFIC TASK OR A LIMITED SET OF TASKS VERY EFFECTIVELY.

By contrast, researchers often speak of General AI (Artificial General Intelligence, AGI). General AI is still theoretical—it has not yet been created. In principle, a General AI would have the ability to learn and apply intelligence across many tasks, just as humans can. It could transfer knowledge from one situation to another, adapt to unfamiliar challenges, and reason flexibly in new contexts. In other words, it would be capable of applying its intelligence to virtually any domain.

GENERAL AI COULD LEARN AND APPLY INTELLIGENCE FLEXIBLY
ACROSS MANY DIFFERENT TASKS.

Narrow AI applications, on the other hand, are typically classified according to the kind of cognitive function they mimic. The five most widely recognized categories are:

1. Perception
2. Language
3. Reasoning and Decision-Making
4. Prediction and Forecasting
5. Control and Robotics

PERCEPTION

Perception is the stage where AI systems handle raw input from the world. Just as humans use their senses to see and hear, AI uses sensors, cameras, and microphones to detect signals. The goal is to transform pixels, sound waves, or other raw data into information the system can work with.

PERCEPTION IS THE ABILITY OF AI TO INTERPRET SENSORY INPUT,
ESPECIALLY VISION AND SOUND.

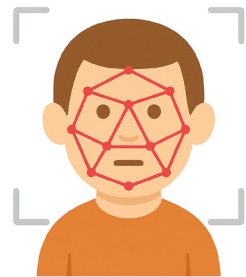
Computer vision

Computer vision allows AI to recognize and analyze visual information. This includes identifying faces in photos, detecting objects in security footage, and even reading handwriting.

For example, in medicine, AI systems are trained to analyze X-rays, CT scans, and MRI images, highlighting areas that might contain signs of disease. Farmers use drones with computer vision to monitor crop health, spotting areas affected by pests or lack of water. Retailers rely on AI-powered cameras to track how shoppers move through stores, helping them redesign layouts to improve shopping experiences. Many smartphones now use facial recognition as a security feature.

Examples of real-world applications:

- Identifying faces in photos
- Detecting objects in security footage
- Reading handwriting
- Analyzing X-rays, CT scans, and MRI images in medicine
- Monitoring crop health with drones in agriculture
- Tracking shopper movement in retail stores
- Smartphone facial recognition for security



Speech recognition

Speech recognition is a branch of AI perception. It allows AI to convert spoken language into digital signals that can later be processed as text. This technology makes it possible for computers to “hear” human speech. Speech recognition itself is not responsible for understanding or acting on the speech, but it provides the raw input that other AI systems require.

For example, systems like Siri, Alexa, and Google Assistant rely on speech recognition to convert spoken words into text before carrying out commands.

Automatic captioning tools also depend on speech recognition to generate subtitles. In some applications, speech recognition works together with language processing, such as in real-time translation between languages.

Examples of real-world applications:

- Virtual assistants (Siri, Alexa, Google Assistant)
- Automatic captioning tools for videos
- Real-time translation apps

**Limitations & Challenges**

Although AI perception has made impressive progress, it does not “see” or “hear” in the way humans do. Instead, it processes raw data such as pixels in an image or sound waves in an audio file. This allows AI systems to recognize patterns quickly and sometimes with remarkable accuracy, but there are important limitations:

- **Task-specific training:** An AI trained to recognize cats in photos cannot automatically recognize dogs without specific training, just as a speech recognition system trained on one language cannot understand another language without retraining. Each new skill requires its own dataset and training process.
- **Lack of common sense:** AI may detect shapes or sounds but does not understand their meaning in context—for example, it might confuse a toy cat with a real one.
- **Data dependence:** Performance depends heavily on the quality and diversity of the data. If the training data is incomplete or biased, the system may make mistakes in real-world situations.
- **Fragility:** Small changes in input, such as unusual lighting in an image or background noise in audio, can cause AI systems to fail where humans would still succeed.

LANGUAGE

Language is one of the most complex human abilities, involving not only words but also meaning, context, and culture. AI has made remarkable progress in this field through Natural Language Processing (NLP), which allows computers to process and generate human language in both written and spoken forms.

NATURAL LANGUAGE PROCESSING ANALYZES, TRANSLATES, AND GENERATES HUMAN LANGUAGE IN TEXT OR SPEECH.

Real-world applications of AI in language include:

- Translation tools (e.g., Google Translate) that enable instant communication across languages.
- Chatbots that provide customer service by analyzing questions and generating responses.
- Large language models (e.g., ChatGPT, Gemini) that can summarize documents, draft essays, or even write computer code.
- Content filtering on social media platforms to detect spam, offensive, or inappropriate material.
- Sentiment analysis that helps businesses understand customer opinions by scanning reviews and posts.
- Educational tools that give feedback on student writing, correcting grammar and suggesting improvements.



In all these applications, Natural Language Processing (NLP) specifically handles the processing of language—analyzing, interpreting, and generating text. These systems often work alongside speech recognition, which first converts spoken words into text before NLP can take over.

Examples of real-world applications:

- Translation tools
- Chatbots
- Large language models
- Content filtering on social media platforms
- Sentiment analysis
- Educational tools

**Limitations & Challenges**

AI language systems can translate text, summarize information, or generate conversations, but they face several challenges:

- Shallow understanding: AI does not truly “understand” meaning; it relies on statistical patterns in data, which can lead to irrelevant or confusing answers.
- Bias in training data: Language models can reproduce stereotypes or offensive language if such patterns are present in the data.
- Difficulty with context: Long conversations, sarcasm, humor, and cultural references are often misunderstood.
- Errors in translation: While real-time translation is powerful, idioms and cultural expressions are frequently mistranslated.

These issues show that AI language tools are valuable helpers but cannot replace human judgment or cultural awareness. They are best used as support systems that speed up tasks, provide suggestions, or make communication easier, while leaving interpretation and final decisions to people. Understanding their limitations helps users apply them wisely and avoid overreliance. In this way, AI language tools can enhance human abilities rather than attempt to replace them.

REASONING AND DECISION-MAKING

Reasoning involves drawing conclusions from information and making choices based on those conclusions. In AI, reasoning and decision-making represent the choice stage, where the system evaluates options and selects an action. While humans rely on experience and common sense, AI applies rules, models, or learned patterns to make choices within a specific domain. Although limited in scope, AI reasoning can be highly effective when focused on a well-defined task.

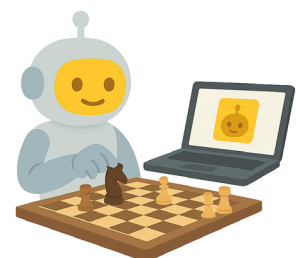
REASONING AND DECISION-MAKING ARE AI'S ABILITY TO EVALUATE OPTIONS AND CHOOSE ACTIONS WITHIN A SPECIFIC DOMAIN.

AI reasoning appears in many areas of daily life. Recommender systems are one of the most common examples. Platforms like YouTube, Netflix, or Spotify analyze a user's past behavior and suggest what to watch or listen to next. Online retailers such as Amazon use similar methods to recommend products based on browsing and purchase history.

In logistics and planning, delivery companies apply AI to calculate the most efficient routes for trucks, balancing distance, fuel, and time. Airlines rely on AI systems to adjust ticket prices depending on demand. Another striking example is in game-playing, where AIs that defeated world champions in chess and Go used reasoning strategies to explore possible moves and select the most promising path.

Examples of real-world applications:

- Recommender systems to suggest videos and music
- Online retailers to recommend products
- Logistics systems to calculate the most efficient route
- Airlines to adjust ticket prices
- Game-playing AIs to choose the best moves



In conclusion, AI reasoning is powerful when applied to narrow, clearly defined problems. It can increase efficiency, improve recommendations, and solve complex challenges in ways that humans alone may find difficult.

Limitations & Challenges

Despite its strengths, AI reasoning faces several important limitations:

- **Domain restriction:** AI can only reason within the data and rules it has been trained on. It cannot transfer decision-making skills to unrelated tasks.
- **Lack of common sense:** Algorithms may choose mathematically optimal solutions that are impractical or unsafe in real life.
- **Hidden bias:** Decisions made by AI in areas such as hiring or lending may reflect biases in the training data, leading to unfair outcomes.
- **Transparency:** Many decision-making systems are “black boxes,” making it difficult to explain how or why a conclusion was reached.

These challenges raise concerns about trust and accountability, especially in areas like finance, healthcare, and law enforcement. In these fields, even small mistakes or hidden biases can cause serious harm. If people cannot see how AI makes its decisions, it is hard to question or fix unfair outcomes. This makes it important to keep AI systems fair, clear, and supervised by humans.

Notes

Common Misconceptions

- *AI can think like humans.*

AI does not understand or think — it recognizes patterns and follows instructions.

- *AI is always right.*

AI can make mistakes, especially if the data is biased, incomplete, or poor quality.

PREDICTION AND FORECASTING

Prediction is one of AI's greatest strengths. By analyzing large amounts of past data, AI can identify patterns and extend them into the future.

PREDICTION AND FORECASTING USE PAST AND CURRENT DATA TO IDENTIFY PATTERNS AND PROJECT FUTURE OUTCOMES.

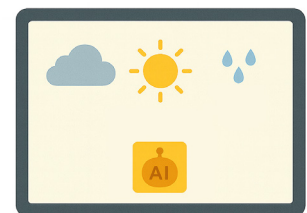
Some applications are now highly established. In finance, AI forecasts stock trends and detects suspicious transactions that may indicate fraud. In healthcare, predictive systems estimate the likelihood of disease outbreaks and assess patient risks. In meteorology, AI improves weather forecasts by analyzing satellite images and atmospheric data. And in retail and e-commerce, predictive tools anticipate customer demand, manage inventory, and reduce waste.

Other applications are still emerging. In sports, teams use AI to analyze player performance and suggest training strategies. In education, predictive analytics helps teachers identify students who may be at risk of falling behind. Governments are also beginning to apply predictive AI in urban planning, such as analyzing traffic flows or planning energy use.

These applications are powerful, yet they also have limits. Predictions are never perfect; their accuracy depends on the quality of the data. For example,

Examples of real-world applications:

- Finance to forecast stock trends
- Healthcare to predict disease outbreak
- Meteorology to generate weather forecasts
- Retail and e-commerce to anticipate demand
- Sports to analyze player performance
- Education to identify students at risk of falling behind
- Governments to predict traffic and energy needs



an AI trained only on sunny weather cannot perform well in a rainy climate. And without human oversight, relying too heavily on predictions can lead to costly mistakes.

Limitations & Challenges

AI prediction relies on patterns in past data, which creates important challenges:

- **Dependence on data quality:** Predictions are only as good as the data used to train the system. If the data is incomplete, biased, or inaccurate, the predictions will also be flawed.
- **Inability to handle the unexpected:** AI struggles with rare or new events that never appeared in its training data, so it may fail in unusual situations.
- **Overconfidence:** Predictions often look exact, but even small errors can grow into big problems, such as in financial trading or weather forecasting.
- **Ethical risks:** Predictive systems in areas like policing or education may unfairly label people, which can limit their opportunities and create long-term harm.

Predictions are powerful tools that can guide decisions, but they should always be checked and supervised by humans.

Notes

Common Misconceptions

- *AI will replace humans completely..*

AI can automate tasks, but it still requires human oversight, creativity, and judgment.

- *Robots and AI are the same thing.*

AI is software (algorithms and models), while robots are machines that may use AI to act.

CONTROL AND ROBOTICS

Control refers to the ability of AI to direct machines that act in the physical world. Robotics combines perception, reasoning, and prediction into coordinated action.

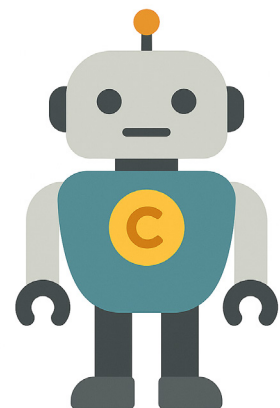
CONTROL AND ROBOTICS IS THE USE OF AI TO DIRECT MACHINES THAT ACT IN THE PHYSICAL WORLD.

One of the most familiar examples of AI in control is the self-driving car. These vehicles use cameras and sensors to take in information about their surroundings. AI perception helps them “see” the road, traffic signs, and obstacles. Prediction is used to guess how other drivers or pedestrians might move, and reasoning helps the car decide when to change speed, stop, or turn.

Robots in industry are another important use of AI. In factories, robotic arms assemble products with speed and precision, often working faster and more accurately than humans on repetitive tasks. Delivery robots and drones also rely on AI to plan safe routes and avoid obstacles, whether they are navigating sidewalks, warehouses, or the air.

Examples of real-world applications:

- Self-driving cars to navigate roads safely
- Industrial robots to assemble products in factories
- Delivery robots to carry goods through warehouses
- Drones to deliver packages
- Robotic surgery to assist doctors with operations
- Agricultural robots to harvest crops and sort produce
- Disaster-response robots to search for survivors in dangerous areas



AI-controlled robots are also becoming more common in other fields. In healthcare, robotic surgery uses AI to guide surgical instruments with great precision, which can shorten recovery times for patients. In agriculture, robots can harvest crops, sort produce, or spray fields, helping farmers reduce waste and save time. In disaster response, AI-driven robots are being tested to search for survivors in collapsed buildings, flooded areas, or other dangerous environments where sending humans would be too risky.

In all these cases, AI is not just analyzing information but also acting in the physical world, turning data into movement and decisions. This shows how AI control and robotics bring together perception, prediction, and reasoning to create systems that can work alongside people in many different settings.

Limitations & Challenges

AI systems that act in the physical world combine perception, reasoning, and prediction, but they are far from perfect:

- **Limited adaptability:** Robots trained for one task, such as picking strawberries, cannot easily adapt to another, such as harvesting apples.
- **Environmental challenges:** Self-driving cars may work well in one city but struggle in different weather, traffic, or road conditions.
- **Mechanical and safety risks:** Hardware failures or incorrect decisions can lead to dangerous accidents.
- **High costs:** Building and maintaining advanced robots is expensive, which limits widespread adoption.

These challenges remind us that while robots and AI-controlled machines can perform repetitive or risky tasks, they still require human supervision and intervention.

ETHICS AND RESPONSIBILITY

As AI becomes more common in daily life, questions of ethics—what is right or wrong—become increasingly important. Ethical considerations guide how AI should be designed, used, and supervised so that it benefits society while avoiding harm.

WHY ETHICS MATTERS

AI is powerful because it extends core human abilities—perception, language, reasoning, prediction, and control—into machines. But power comes with responsibility. Without ethical guidelines, AI could be misused, cause harm, or increase inequality.

For example, if a healthcare AI is trained only on data from adults, it may give poor advice when used with children, creating unfair outcomes. A predictive policing system that relies on biased crime data may target certain communities more than others, reinforcing social injustice.

Ethics ensures that AI systems are designed and used in ways that respect human rights and dignity. It reminds us to ask questions such as: *Who benefits from this technology? Who might be harmed? How can risks be reduced?*

Ethical considerations also help build trust. If people believe that AI is fair, transparent, and accountable, they are more likely to accept and use it. If they do not, society may reject valuable technologies out of fear or suspicion.

In short, ethics matters because AI does not exist in isolation—it affects real people, influences real decisions, and shapes the future of communities worldwide. By applying ethical principles, we can guide AI to support fairness, safety, and positive social change.



UNETHICAL CASE STUDIES

- ***Bias and Fairness***



Case Study: A university used an AI system to help rank student applications. Because the system was trained mostly on past data from high-income schools, it gave unfair preference to wealthy students while disadvantaging equally qualified students from less privileged backgrounds. This shows how bias in data can lead to unfair results.

- ***Privacy***



Case Study: A major social media company secretly used AI to scan private messages in order to target users with ads. Many people felt their personal conversations had been monitored without consent, which caused public backlash and loss of trust. This highlights why privacy is a key ethical concern in AI.

- ***Transparency and Accountability***



Case Study: In the U.S., some courts used an AI tool called COMPAS to predict how likely defendants were to reoffend. Judges used these scores in bail, parole, and sentencing decisions. Researchers later found the system gave harsher scores to Black defendants than to white defendants, and because its reasoning was hidden, unfair outcomes were hard to challenge.

- ***Dependence and Displacement***



Case Study: A factory introduced AI-controlled robots to increase production speed. While profits rose, hundreds of workers lost their jobs without retraining opportunities. This created unemployment and hardship in the local community, showing how AI can displace workers if changes are not managed responsibly.

SUMMARY

Narrow AI vs. General AI

- **Narrow AI:** Current real-world AI; designed for specific tasks (e.g., translation, face recognition). Cannot transfer skills to new tasks.
- **General AI:** A future goal; would learn and apply intelligence across many tasks like a human. Not yet achieved.

Five Cognitive Categories of AI

- **Perception (Input stage):** Converts raw data (images, sounds) into usable information. Examples: computer vision, speech recognition.
- **Language (Processing stage):** Interprets and generates human language. Examples: translation tools, chatbots, content filtering.
- **Reasoning & Decision-Making (Choice stage):** Evaluates options and selects actions. Examples: recommender systems, logistics planning, dynamic pricing.
- **Prediction & Forecasting (Future stage):** Uses past data to estimate future outcomes. Examples: stock forecasts, weather models, education risk alerts.
- **Control & Robotics (Action stage):** Directs machines to act in the physical world. Examples: self-driving cars, industrial robots, robotic surgery.

Limitations & Challenges

- **Perception** – Task-specific training, lack of common sense, data dependence, fragility
- **Language** – Shallow understanding, bias in data, context difficulty, translation errors
- **Reasoning and Decision-Making** – Domain restriction, lack of common sense, hidden bias, transparency issues
- **Prediction and Forecasting** – Data dependence, unexpected events, overconfidence, ethical risks
- **Control and Robotics** – Limited adaptability, environmental challenges, safety risks, high costs

Ethics and Responsibility

- **Bias and Fairness:** AI can reproduce social inequalities if trained on biased data.
- **Privacy:** AI may misuse personal data without consent.
- **Transparency and Accountability:** “Black box” systems make it hard to explain or challenge decisions.
- **Dependence and Displacement:** Overuse of AI can replace jobs and reduce human skills.
- **Why Ethics Matters:** Ensures AI respects human rights, builds trust, and supports fairness, safety, and positive social change.

Activity 4.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Narrow AI can only perform specific tasks and cannot transfer skills to other tasks.
<u>F</u>	General AI has already been achieved and is widely used in real-world applications.
<u>T</u>	Perception in AI refers to converting raw data like images and sounds into usable information.
<u>F</u>	Language AI is mainly about understanding human emotions through language.
<u>T</u>	Reasoning and Decision-Making AI evaluates options and selects actions, such as in recommender systems and logistics planning.
<u>T</u>	Prediction and Forecasting AI uses past data to estimate future outcomes, such as stock trends or weather forecasts.
<u>F</u>	Control and Robotics represents the input stage of AI, focusing on data collection from sensors.
<u>T</u>	One limitation of AI perception is fragility, meaning small changes in lighting or noise can cause errors.
<u>F</u>	AI systems are always transparent and easy to explain, which ensures accountability.
<u>T</u>	Ethical concerns include bias and fairness, privacy, transparency, and the risk of job displacement.

Activity 4.2

Answer the questions below.

1. Which of the following best describes Narrow AI?
 - a) AI that can learn any task like a human
 - b) **AI that can only perform specific tasks**
 - c) AI that always understands meaning
 - d) AI that replaces human judgment completely
2. What is General AI?
 - a) Current real-world AI used in translation tools
 - b) **A future AI that could learn and apply skills across many tasks**
 - c) An AI limited to perception and prediction only
 - d) An AI already used in self-driving cars
3. Which cognitive category involves converting raw data like pixels and sound waves into usable information?
 - a) **Perception**
 - b) Prediction and Forecasting
 - c) Reasoning and Decision-Making
 - d) Language
4. Which is an example of AI in Language processing?
 - a) Stock forecasting
 - b) Self-driving cars
 - c) **Translation tools**
 - d) Robotic surgery
5. AI used in recommender systems, logistics planning, and dynamic pricing belongs to which cognitive category?
 - a) Perception
 - b) **Reasoning and Decision-Making**
 - c) Control and Robotics
 - d) Language

 Activity 4.2 (continued)

Answer the questions below.

6. What does Prediction and Forecasting AI mainly do?
 - a) Uses past data to estimate future outcomes
 - b) Summarizes large documents
 - c) Converts sound waves into text
 - d) Decides which move to play in a game
7. Which example belongs to Control and Robotics?
 - a) Detecting faces in photos
 - b) Chatbots for customer service
 - c) Weather forecasting
 - d) Self-driving cars
8. Which is a limitation of AI perception?
 - a) Errors in translation
 - b) Fragility to small changes
 - c) Hidden bias in hiring
 - d) Overconfidence in forecasts
9. Why is transparency and accountability an ethical concern?
 - a) AI always explains its decisions
 - b) AI systems are inexpensive and adaptable
 - c) AI never makes mistakes if data is good
 - d) “Black box” systems make it hard to explain or challenge outcomes
10. Which of the following is NOT listed as an ethical concern of AI?
 - a) Bias and fairness
 - b) Privacy
 - c) Job displacement
 - d) Faster internet speeds

Activity 4.3

Read the narrative and answer the questions below.

- a. A new security system at a train station uses AI cameras to detect suspicious items. One morning, a child leaves behind a toy water gun. The AI system quickly flags it as a potential weapon, causing the station to be evacuated. Later, officials realize it was only a harmless toy.

1. Which cognitive function of AI is being used in this scenario?

Perception – analyzing visual input from cameras.

2. Why did the system mistake the toy water gun for a real weapon?

It recognized the shape but not the context.

3. What limitation of AI perception does this scenario highlight?

AI perception lacks common sense.

4. How could human oversight have prevented the unnecessary evacuation?

A human officer could have reviewed the flagged item and recognized it was only a toy before ordering an evacuation.

- b. An international company uses an AI translation tool for customer support. One customer in America writes, “It’s raining cats and dogs here, will that make my delivery late?” The AI translates the phrase literally into Mandarin. The customer support person reads it and is confused: “How could it rain cats and dogs?!”

1. Which cognitive function of AI is being used in this scenario?

Language – processing and translating text.

2. Why did the translation confuse the customer support person?

The AI translated the idiom word-for-word.

3. What limitation of AI language does this scenario highlight?

AI language has trouble with context and idiomatic expressions.

4. How could the company improve the accuracy of its translations?

By using human review or improving the system with more cultural/linguistic data.

Activity 4.3 (continued)

Read the narrative and answer the questions below.

- c. A food delivery company uses AI to assign drivers to orders. Drivers notice that short, high-paying deliveries mostly go to certain neighborhoods, while other areas get fewer and less profitable orders. The company later admits the AI was trained on past data that favored wealthier neighborhoods.

1. Which cognitive function of AI is being used in this scenario?

Reasoning and Decision-Making – assigning deliveries.

2. What unfair pattern did the drivers notice in the AI's decisions?

Wealthier neighborhoods got more profitable orders.

3. What limitation of AI reasoning does this scenario highlight?

Hidden bias in the training data.

4. How might the company fix this problem?

Retrain with fairer data and set rules for equal distribution.

- d. A city uses AI to forecast traffic for road repairs. The system predicts light traffic for a weekend, even though a major sports event is scheduled. Because the AI had never seen such an event before, it ignored it. Human planners assumed the AI accounted for everything, so construction went ahead. On game day, thousands of fans arrived, and the city was gridlocked.

1. Which cognitive function of AI is being used in this scenario?

Prediction and Forecasting – estimating future traffic.

2. Why did the prediction fail, even though the AI had data about the sports event?

It had no prior examples of such an event, so it ignored the data.

3. What limitation of AI prediction does this highlight?

Inability to handle the unexpected.

4. How could humans have helped avoid the problem?

By being aware of AI's limits, the planners should have reviewed the forecast critically and applied human judgment for unusual events.

Activity 4.3 (continued)

Read the narrative and answer the questions below.

- e. A city tests self-driving buses on public roads. Most of the time, the buses run smoothly. One day, however, a pedestrian suddenly crosses the street outside a crosswalk. The bus hesitates for a moment before stopping, narrowly avoiding an accident. City officials realize the AI system was not fully prepared for unpredictable human behavior.

1. Which cognitive function of AI is being used in this scenario?

Control and Robotics – managing self-driving buses.

2. What nearly caused the accident?

The AI hesitated when faced with unexpected pedestrian behavior.

3. What limitation of AI control and robotics does this highlight?

Safety risks – AI-controlled machines may fail in unpredictable real-world situations.

4. How could humans help reduce this type of risk?

By being aware of AI's limits and ensuring human oversight or emergency controls are in place.

- f. A tech company announces a new AI that can drive a car, write novels, prepare food, and give medical advice—all without retraining. Many people line up to buy this amazing new technology. Would you buy it, or would you be skeptical?

1. What type of AI is being claimed in this scenario?

General AI – an AI that can perform many different tasks across domains.

2. Why might people be excited about such a system?

Because it promises to combine many human-like abilities into one system.

3. Why should we be skeptical of this claim today?

We should be skeptical because General AI has not yet been achieved; all existing AI is still Narrow AI.



Activity 4.4 - Exploring AI in the Real World

In this activity, you will interact with an AI which will act as a mentor, providing examples, questions, and explanations to deepen your understanding of AI applications, limitations, and ethics. In the shared folder you will find separate text documents, each containing the prompt you need for the activity, as well as the PDF of the lesson summary.

Access the folder from: bit.ly/C4_AI_Activity4_4

a. Scenario Exploration

In this part, you will practice identifying AI cognitive functions and their limitations through real-world scenarios.

- Perception
- Language
- Reasoning & Decision-Making
- Prediction & Forecasting
- Control & Robotics

Steps to Follow:

1. Open the shared folder and download the PDF of the lesson summary.
2. Attach the PDF in ChatGPT (or another LLM).
3. Open the Scenario Exploration file, then copy the full prompt and paste it into ChatGPT.
4. You will receive one short story scenario at a time, each showing a real-world application of AI in one of the five cognitive functions.
5. Each scenario hides a limitation or challenge for you to uncover.
6. After the scenario, you will be asked four comprehension questions, one at a time. Answer each question, and the AI will tell you if you are correct, explain briefly, and correct you if necessary.
7. At the end of each scenario, the AI will review the questions you got wrong, highlight key topics you struggled with, and suggest one follow-up exercise.
8. The next scenario will move on to a different cognitive function (no repeats). Continue until all five functions are covered.

Always explain your reasoning, even if unsure — that is what the AI will evaluate.

Don't rush.

b. Questions & Answers

In this part, you will test your understanding of the lesson using the PDF provided in the folder. You can choose between:

- True/False Questions
- Multiple-Choice Questions

Each activity has its own text file with a prompt.

Steps to Follow:

1. Open the shared folder and download the PDF of the lesson summary.
2. Attach the PDF in ChatGPT.
3. Copy the prompt for the activity you want to practice (True/False or Multiple Choice) and paste it into ChatGPT.
4. The AI will create 10 questions based on the summary:
 - If you chose True/False, each question will be answered with “True” or “False.”.
 - If you chose Multiple Choice, each question will have 4 options (A, B, C, D).
5. For each question:
 - Read carefully and submit your answer.
 - The AI will tell you if you are correct, provide the correct answer if needed, and give a short explanation.
6. After all questions:
 - The AI will list the questions you got wrong.
 - It will give you a short review of the key topics you struggled with.
 - It will suggest a follow-up exercise to strengthen your understanding.

Tips:

- For True/False: think about why the statement is true or false before answering.
- For Multiple Choice: read all four options carefully before choosing.
- Use the AI’s feedback to focus on the concepts you find most difficult.
- Repeat the activity for more practice — the AI can generate new sets of questions each time.

c. Ethics Debate

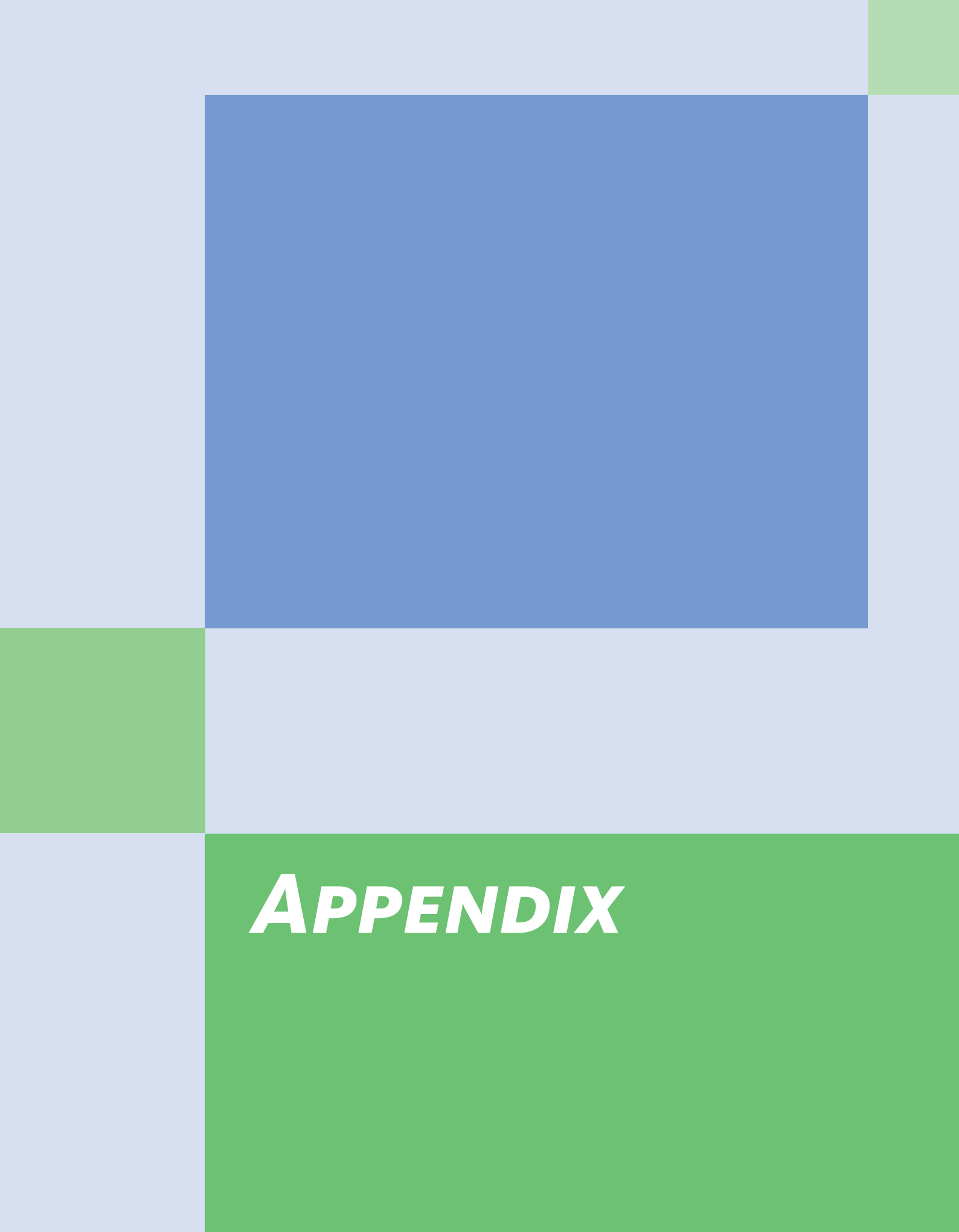
This optional part lets you practice thinking critically about AI and society.

Steps to Follow:

1. Copy the Ethics Debate Prompt into ChatGPT.
2. The AI will first list the available debate themes for you to choose from: Bias, Privacy, Transparency, Job Displacement.
3. After you select a theme, the AI will give you one ethical scenario for it.
4. You will then be asked 3–4 debate questions, one at a time. After each answer, the AI will tell you if your response is strong or weak, challenge you with a counterargument, and suggest improvements.
5. At the end of the debate, the AI will give you a short review of the ethical concepts and suggest one follow-up exercise.
6. Finally, the AI will ask if you want to try another theme. You may continue until you've practiced as many themes as you like.

Tips:

- There's no single "right" answer. Focus on reasoning clearly.
- Use evidence from the lesson (bias, fairness, human oversight).

The background of the page is a light blue grid. A large blue rectangle is positioned in the upper left quadrant. A green rectangle is in the upper right corner. A green rectangle is on the left side, below the blue rectangle. A large green rectangle occupies the bottom right portion of the page, containing the word 'APPENDIX' in white.

APPENDIX

GLOSSARY

TERM	DEFINITION	PAGE
Accuracy	A measure of how often a model's predictions are correct, calculated as the percentage of correct predictions out of the total.	p. 177
Actionable Information	Information that has been interpreted and can be directly used to make informed decisions.	p. 21
Adjacency List	A way to represent a graph by listing each node and the nodes it connects to, along with edge information.	p. 32
Algorithm	A step-by-step set of instructions that tells a computer how to solve a problem or perform a task. Algorithms can manage data, search through it, or arrange it in order.	p. 33, 68, 139
Array	A linear data structure where items are stored one after another in a single line. Arrays have a fixed size and usually hold items of the same type, so new items cannot be added or removed once created.	p. 12, 33
Artificial General Intelligence (AGI / General AI)	A theoretical form of AI that could learn and apply intelligence across many different tasks, like a human.	p. 217
Artificial Intelligence (AI)	The field of computer science focused on building machines that can perform tasks that normally require human intelligence.	p. 159
Ascending Order	Sorting data from smallest to largest or from A to Z.	p. 95
Association	A link formed between two ideas, such as connecting the image of a cup with the word "cup."	p. 158
Bar Chart	A chart that represents values using vertical or horizontal bars, useful for comparing quantities across categories.	p. 15
Best-Case Scenario	The situation where an algorithm finishes its task in the fewest possible steps (e.g., finding the item on the first try).	p. 73
Bias	An unfair pattern in AI predictions or decisions caused by biased training data or flawed design.	p. 228
Big Data	Large and varied sets of data that help AI models learn better and make more reliable predictions.	p. 184
Binary Search	A search algorithm for sorted datasets that repeatedly divides the list in half to find the target more quickly.	p. 75
Black Box	A system, often in AI, where the decision-making process is hidden and hard to explain or challenge.	p. 223
Block (in programming)	A section of code grouped together to perform a task; examples include initialization, main body, and termination.	p. 140
Bubble Sort	A sorting algorithm that repeatedly compares and swaps adjacent items until the list is sorted.	p. 111

Categorical Data	Another term for qualitative data; data that can be grouped and labeled but not measured.	p. 18
Categorical Prediction	A type of prediction where the AI assigns inputs to categories, such as identifying if an object is a bottle or not.	p. 175
Chart	A visual method of presenting data so patterns and trends can be easily seen (e.g., line, bar, pie, scatter).	p. 14
Child Node	A node in a tree that descends from another node (its parent).	p. 46
Cleaning Data	The process of removing errors or incorrect entries from a dataset so the data is accurate.	p. 12
Compare, Shift, Insert	The basic strategy used by insertion sort: compare items, shift larger ones, and insert the held item in the right place.	p. 99
Continuous Data	Quantitative data that can take any value within a range, such as temperature or time.	p. 16
Continuous Learning	The idea that a model keeps improving over time as it encounters new data and situations.	p. 165
Continuous Prediction	A type of prediction where the AI estimates a numerical value, such as house prices or temperatures.	p. 178
Control & Robotics	The use of AI to direct machines in the physical world, such as self-driving cars or delivery robots.	p. 226
Criterion (Sorting Criterion)	The rule or property used to decide how to sort data, such as by score, name, or price.	p. 95
Data	Raw input that has not yet been processed to add meaning.	p. 8
Data Structure	The specific way in which data is arranged and stored in a computer.	p. 12
Decision-Making	The process of selecting the best choice from several options by comparing their outcomes and choosing the one with the highest utility.	p. 197
Dependence & Displacement	When overreliance on AI reduces human roles or causes job loss without proper support or retraining.	p. 228
Descending Order	Sorting data from largest to smallest or from Z to A.	p. 95
Discrete Data	Quantitative data that is countable and can only take whole number values, such as number of students.	p. 17
Dynamic Size	A feature of some data structures (like lists, stacks, and queues) that allows them to grow or shrink as items are added or removed.	p. 34
Edge	In a graph, a link between two nodes that can carry extra information, such as distance or time.	p. 50
Error	The difference between the predicted value and the actual value.	p. 180
Ethics (in AI)	Principles that guide AI to be fair, safe, and beneficial to society while avoiding harm.	p. 228

Execution Time	The amount of time it takes for a program or algorithm to complete its run. It is often used to compare how efficiently different algorithms perform.	p. 72, 124
Explanation	A form of reasoning that works backward from an outcome (a given) to find the most plausible cause.	p. 201
FIFO (First In, First Out)	A principle where the first item added is the first one removed, used in queues.	p. 42
Fixed Size	A feature of some data structures (like arrays and tuples) that means their size cannot change once created.	p. 33
Flow (of a program)	The sequence in which steps of an algorithm or program are carried out.	p. 139
Flowchart	A diagram that shows the steps of a process using symbols and arrows to indicate decisions and actions.	p. 69
General AI	See Artificial General Intelligence (AGI).	p. 217
Generalization	The ability of a model to apply what it has learned to correctly identify or handle new, unseen examples.	p. 163, 180
Given	The starting point of explanation, an accepted observation or outcome that needs to be explained.	p. 201
Graph	A non-linear data structure where nodes are connected by edges, allowing many-to-many relationships.	p. 32
Hierarchical Structure	An arrangement where items are organized in levels, like in a tree with parent and child nodes.	p. 31
Indentation	The spacing used in code to show which instructions belong to a block or loop.	p. 140
Index	The position number of an item in a list or array, usually starting at 0. It helps locate and access items in a linear data structure.	p. 35, 70
Inference	A form of reasoning that links known facts, observations, or predictions to reach a new conclusion.	p. 195
Information	Data that has been processed and given meaning, making it useful for understanding or decision-making.	p. 8
Inner Loop	The loop inside another loop; in sorting, it compares and shifts or swaps data items repeatedly during each pass.	p. 104, 117
Insertion Sort	A sorting algorithm that builds a sorted section one item at a time by comparing, shifting, and inserting.	p. 99
Interpreting Data	The step of explaining what data means, asking “why” it happened and “what should we do.”	p. 20–21
Knowledge Base	The collection of rules, facts, or learned patterns that reasoning relies on to connect conditions to outcomes.	p. 206
Labeling Data	Adding descriptive tags or identifiers (like “temperature in °C”) to clarify what the data represents.	p. 11

Language (AI)	AI systems that process, understand, or generate human language in text or speech.	p. 220
Leaf Node	A node in a tree that has no children.	p. 46
Learning	The ability to form new associations and make use of them through repetition and pattern recognition.	p. 158
LIFO (Last In, First Out)	A principle where the last item added is the first one removed, used in stacks.	p. 39
Line Chart	A chart where data points are connected by a line, often used to show trends over time.	p. 14
Linear Data Structure	A type of data structure where items are arranged sequentially, like arrays, lists, stacks, and queues.	p. 30
Linear Search	A simple search algorithm that checks each item in a list one by one until it finds the target or reaches the end.	p. 68
List	A linear data structure with a dynamic size that can store mixed data types and allows insertion and removal at any position.	p. 36
Logarithmic Growth	A type of growth where progress increases quickly at first but slows as the dataset gets larger (common in binary search).	p. 80–81
Logic	The rules that guide reasoning and ensure conclusions follow consistently from premises.	p. 194
Loop	A control structure that repeats a set of instructions until a condition is met.	p. 104
Machine Learning	A branch of AI where a computer learns from data by recognizing patterns and forming associations.	p. 159
Mean Absolute Error (MAE)	A way to measure how accurate a model's numerical predictions are, by calculating the average size of the errors (ignoring whether they are positive or negative).	p. 180–181
Memory Latency	The delay between requesting data from memory and receiving it.	p. 140
Misclassification	A prediction error where a model gives the wrong output, such as failing to recognize a cup with a handle or labeling a tomato as an apple.	p. 163, 177
Model	The AI's internal system that stores recognized patterns and associations, used to make predictions or classifications.	p. 162
Narrow AI	AI designed for a specific task or limited set of tasks, such as translation or facial recognition.	p. 216
Natural Language Processing (NLP)	A branch of AI that analyzes, interprets, and generates human language.	p. 220
Node	A unit of data in tree or graph structures that can connect to other nodes.	p. 46
Non-Linear Data Structure	A type of data structure where data items are connected in more complex ways, like trees and graphs.	p. 31

Organizing Data	Arranging data in ways such as cleaning, structuring, or sorting so it becomes easier to analyze.	p. 11
Outer Loop	The main loop that controls how many passes through the list an algorithm makes.	p. 105, 118
Pair, Compare, Swap	The basic strategy used by bubble sort: compare pairs of adjacent items and swap them if needed.	p. 111
Parent Node	A node in a tree that has one or more child nodes.	p. 46
Pattern Recognition	The ability of a system to detect recurring regularities, such as shapes or sounds that repeat.	p. 160
Perception (AI)	The ability of AI to interpret sensory input, like vision or sound, using sensors, cameras, or microphones.	p. 217
Performance	A measure of how efficiently an algorithm runs, often judged by the number of steps it takes and the time it needs to finish.	p. 72, 124
Pie Chart	A circular chart divided into slices that represent proportions of a whole.	p. 15
Plausibility	A measure of how believable or reasonable an explanation is, based on how well its premises fit the outcome.	p. 201
Prediction	When an AI system uses its model to estimate the most probable outcome in a new situation.	p. 174
Prediction & Forecasting (AI)	Using past and current data to identify patterns and estimate future outcomes.	p. 224
Premise	The starting point of reasoning, which can be a fact, observation, or prediction.	p. 195
Privacy	The right to keep personal data protected from misuse or unauthorized access.	p. 228
Qualitative Data	Descriptive data that can be categorized into groups (like colors or gender) but not measured.	p. 18
Quantitative Data	Numerical data that can be measured or counted.	p. 16
Queue	A linear data structure with a dynamic size where items are added at the end and removed from the beginning, following FIFO.	p. 42
Reasoning	A structured way of thinking that helps us discover, decide, or explain by following logical rules.	p. 194
Reasoning & Decision-Making (AI)	AI systems that evaluate options and choose actions within a defined area, like recommender systems.	p. 222
Root Node	The top node in a tree, from which all other nodes descend.	p. 31, 46
Scalability	The ability of an algorithm or AI system to keep performing well as the dataset grows larger or the tasks become more complex.	p. 130, 225
Scatter Plot	A chart that shows the relationship between two continuous variables as individual points.	p. 14
Search Index	The current position in a dataset that an algorithm is checking.	p. 78

Search Range	The section of a dataset that an algorithm is currently looking at during a search.	p. 78
Sentiment Analysis	An AI tool that scans reviews, posts, or text to detect opinions, such as positive, negative, or neutral.	p. 220
Sorted List	A list in which items are arranged in order (such as numbers from smallest to largest or words alphabetically).	p. 75
Sorting	The process of arranging data in a specific order based on a chosen criterion.	p. 94
Sorting Data	Re-arranging data according to a criterion such as alphabetical order, numerical order, or date.	p. 13
Speech Recognition	An AI capability that converts spoken words into text.	p. 218
Stack	A linear data structure with a dynamic size where items can only be added or removed from the end, following LIFO.	p. 39
Step Count	The number of individual operations an algorithm performs while completing its task.	p. 124
Structuring Data	Storing data in an organized way, such as in lists (arrays) or tables.	p. 12
Supervised Learning	A type of machine learning where the system is trained with labeled data (e.g., images tagged as “cup”).	p. 160
Swap	An operation where two data items exchange positions, usually requiring three moves in memory.	p. 135
Table	A way of structuring data using rows and columns to handle multiple sets of related data.	p. 12–13
Target	The specific item an algorithm is trying to find in a dataset.	p. 70
Termination	The end block of a program or algorithm that signals completion of the task.	p. 110, 123
Training	The process of improving a model using new data and correcting mistakes.	p. 163
Training Data	The examples used to teach an AI model, helping it learn patterns for making predictions.	p. 178
Transparency & Accountability	The principle that AI systems should explain their decisions clearly so they can be reviewed and corrected if needed.	p. 228
Tree	A non-linear data structure that organizes items hierarchically in nodes connected as parents and children.	p. 31
Tuple	A linear data structure with a fixed size and format that can hold different types of data, often pairing related data together.	p. 37
Unsupervised Learning	A type of machine learning where the system groups data based on similarities without using labels.	p. 161
Utility	A measure of how useful, valuable, or goal-fitting an outcome is, used in decision-making.	p. 200
Variable	A label in programming that holds data and whose value can change.	p. 103

Variety (in data)	Having many different types of examples in the training data, which helps the model make better predictions in new situations.	p. 184
Visualizing Data	Presenting data in a visual format like charts or graphs to make trends and patterns clearer.	p. 14
Volume (in data)	The total amount of training data available to the model, which helps it learn broader patterns.	p. 184
Weight	Extra information stored on a graph edge, such as the distance or time between two nodes.	p. 51
Worst-Case Scenario	The situation where an algorithm takes the maximum possible steps to complete a task (e.g., checking the whole list).	p. 73

INDEX

Accuracy – Measure of performance for categorical predictions; percentage of correct predictions out of total attempts (p.176–178).

Adjacency list – A way of representing a graph by listing each node along with the nodes it connects to and the weight of each edge (p.32, 50–52).

Algorithm – A series of steps designed to perform a specific task, such as sorting a list (p.139–140).

- Algorithm performance – Evaluation of how efficiently an algorithm runs, often measured by execution time in best-case and worst-case scenarios (p.72–74).
- Algorithms (data structures) – Sets of instructions that control data structures, defining rules for size, data type, and operations (p.33).

Artificial General Intelligence (AGI / General AI)

– A theoretical AI system that could learn and apply intelligence flexibly across many tasks, similar to humans. Not yet achieved (p.217, 236).

Associations – Links formed between known and new information; essential for learning (p.158–159).

Array (List) – A linear data structure that stores items sequentially in memory. Arrays have a fixed size and homogeneous data type: items can be updated but not inserted or removed (p.12–13, 33–35).

Bar chart – A chart in which categories are represented by bars; useful for comparing quantities across categories. Can be vertical or horizontal (p.15, 17, 19).

Bias – An ethical concern where AI reproduces unfair patterns from biased training data, leading to unequal treatment (p.228–229).

Best-case scenario – The situation in which an algorithm finds the target immediately or completes its task with the fewest possible steps. For example, in search, it may succeed on the very first item checked (p.73, 85).

Big Data – Large datasets characterized by volume and variety; improve AI models by helping them generalize

and make more accurate predictions (p.184).

Binary search – Search algorithm for sorted datasets that repeatedly divides the search range in half until the target is found or eliminated (p.75–79).

- Flowchart – Diagram showing binary search process (p.77).
- Pseudocode – Step-by-step outline of binary search (p.79).
- Execution time – Grows logarithmically with dataset size, modeled with $\log_2(n)$ (p.80–82).
- Requirements – Dataset must be sorted (p.75, 86).

Black box problem – A limitation of AI reasoning and decision-making where systems lack transparency, making it difficult to explain or challenge outcomes (p.223).

Block – A segment of code that performs a task, usually part of initialization, main body, or termination (p.140).

Bubble sort – Sorting algorithm using the strategy of pair, compare, and swap. Repeatedly passes through the list, swapping adjacent items if out of order (p.111–113, 117–123).

- Process – Uses outer loop (repeats passes) and inner loop (compares/swaps adjacent pairs) (p.117–123).
- Pseudocode – Instructions for bubble sort (p.120–123, 153).
- Performance – Requires more steps than insertion sort; poor scalability; inefficient for large datasets (p.134–138).

Categorical data – Another name for qualitative data, which can be grouped and labeled but not measured (p.18).

Categorical prediction – Prediction task where the AI classifies inputs into categories (e.g., cat or not a cat, language detection) (p.175–176).

Charts – Visual tools to present data. Four common types: line chart, scatter plot, bar chart, and pie chart

(p.14–15).

Child node – A node that descends from a parent node in a tree structure (p.46).

Cleaning data – Removing erroneous or implausible data values before processing (p.12).

Continuous data – Quantitative data that can take any value within a range (e.g., temperature, height, time). Best visualized with line charts or scatter plots (p.16–17).

Continuous learning – The idea that learning does not stop after training; models can keep improving with new data and feedback (p.165).

Continuous prediction – Prediction task where the AI estimates a numerical value (e.g., house prices, rainfall, travel times) (p.178–180).

Control and Robotics – The use of AI to direct machines acting in the physical world. Applications include self-driving cars, industrial robots, delivery drones, robotic surgery, agricultural robots, and disaster-response robots (p.226–227).

Criterion (sorting) – The basis for sorting, such as numeric (scores, prices, time) or alphanumeric (names, book titles) (p.95).

Data – Raw values or signals that can be processed into meaningful information (p.8).

Data structures – Methods of organizing data in computers, such as lists (arrays), tables, stacks, queues, trees, and graphs (p.12–13).

- Dynamic size – Property of structures like lists, stacks, and queues that allows them to grow or shrink as items are added or removed (p.34, 39, 42).

Decision-making – The process of selecting the best choice from several options by weighing their outcomes and choosing the one with the highest utility (p.197–200).

Discrete data – Quantitative data that can only take whole-number values (e.g., number of students). Often visualized using bar charts (p.17).

Edge – A link between two nodes in a graph, often carrying additional information such as distance or latency (p.32, 50–52).

Error – The difference between predicted and actual values; can be positive (underestimation) or negative (overestimation) (p.180).

Ethics and responsibility – Principles ensuring AI respects human rights, fairness, safety, privacy, and accountability, while reducing risks of harm or inequality (p.228).

Execution time – The time required for a program to complete its run.

- In search – Calculated as time per attempt $\times$ number of attempts (p.72, 80–82).
- In sorting – Influenced by step count and memory latency (p.124–125, 131–133, 138).

Explanation – Reasoning backward from an observed outcome to uncover its most plausible causes, contributors, or relationships (p.201–204).

FIFO (First In, First Out) – Principle governing queues; the first item added is the first removed (p.42–44).

Flow & control – Flow is the sequence of steps in a program; control refers to conditions that determine which steps are executed, repeated, or skipped (p.139).

General AI – see Artificial General Intelligence (AGI).

Given – An accepted observation or outcome that serves as the starting point for explanation (p.201–202).

Graph – A non-linear data structure where nodes can connect freely to represent complex relationships. Edges may store additional information (p.32, 49–54).

Hierarchy – Organizational model in which data items are arranged at levels, such as in a tree structure (p.31, 45–47).

Hold index / hold data – Variables in sorting algorithms representing the index and the value temporarily stored for comparison (p.103, 116).

Information – Data that has been processed into meaningful form, typically through identification, organization, visualization, and interpretation (p.8–11, 21).

- Interpreting data – The final step; answering “why” and “what should we do?” to make data actionable (p.20–21).

- Labeling data – Adding descriptive identifiers (e.g., “temperature, outdoor, Celsius”) to clarify meaning and context (p.11).
- Organizing data – Arranging data to make it usable, through cleaning, structuring, or sorting (p.11–13).

Inference – Arriving at a new piece of information by linking premises (facts, observations, or predictions) into a conclusion (p.195–197).

Initialization – The first block of a program, where variables are declared and initial values assigned (p.106–107, 119–120).

Insertion sort – Sorting algorithm using the strategy of compare, shift, and insert. Builds a sorted section one item at a time (p.99–106, 139–140).

- Process – Uses outer loop (moves through each index) and inner loop (compares/shifts until correct position is found) (p.103–106).
- Pseudocode – Instructions for insertion sort (p.106–110, 152).
- Performance – Fewer steps and faster than bubble sort; still scales poorly for very large datasets (p.124–130).

Job displacement – Risk of AI replacing human workers without retraining opportunities, leading to unemployment and inequality (p.229).

Knowledge base – The collection of rules, facts, and learned patterns that reasoning processes rely on to connect conditions to outcomes (p.201).

Labeling data – see Information.

Language (AI cognitive function) – Natural Language Processing (NLP) enables AI to analyze, interpret, and generate human language. Applications include translation tools, chatbots, large language models, sentiment analysis, and educational tools (p.220–221).

Leaf node – A node in a tree with no child nodes (p.46).

Learning – The ability to form and use new associations; requires pattern recognition, association, and repetition (p.158–159, 166).

LIFO (Last In, First Out) – Principle governing stacks; the last item added is the first removed (p.39–41).

Linear search – A simple search algorithm that checks each item in order until the target is found or the end is reached (p.68–71).

- Flowchart – Diagram showing linear search process (p.69–70).
- Pseudocode – Step-by-step outline of linear search (p.71).
- Execution time – Grows linearly with dataset size (p.72–73).
- Applicability – Works on both sorted and unsorted datasets (p.75, 84).

Linear structures – Data structures where items are stored sequentially: array, list, tuple, stack, queue (p.33–34).

List – A linear data structure with dynamic size, accepts mixed data types, and allows insertion or removal at any position (p.36–37).

Logic – The rules that guide reasoning from premises to conclusions; the framework for inference, decision-making, and explanation (p.194).

Logarithmic growth – Growth pattern of binary search execution time; efficiency improves as dataset size increases (p.80–82).

Loop – Control structure used to repeat a set of instructions. Sorting algorithms use outer and inner loops (p.104–106, 117–122).

Machine learning – A branch of AI where computers are designed with learning capabilities to identify patterns, form associations, and improve through repetition (p.159).

Mean Absolute Error (MAE) – A measure of performance for continuous predictions; the average size of errors regardless of direction (p.180–183).

Memory latency – Delay between requesting data from memory and receiving it; becomes significant for large datasets stored outside cache (p.131–133, 138–139).

Misclassification – An error where a model incorrectly labels or classifies input data (p.163–164, 176–177).

Model – The internal system that represents recognized patterns, labels, and rules about what features matter or don’t (p.162–163).

Natural Language Processing (NLP) – AI technology that processes, translates, and generates human language in text or speech (p.220).

Narrow AI – Current form of AI designed to perform a specific task or a small set of tasks very effectively (p.216–217).

Node – A data construct used in trees and graphs to store both data and links to other nodes (p.46–48, 50).

Order (sorting) – Direction in which data is arranged, e.g., ascending (least to greatest) or descending (greatest to least) (p.95).

Parent node – A node with child nodes branching from it in a tree structure (p.46).

Pattern recognition – The ability to detect recurring regularities, such as the shared shape of cups regardless of color or design (p.160).

Pie chart – A circular chart showing categories as slices of a whole, indicating proportions or percentages (p.15, 20).

Plausibility – A measure of how reasonable or believable an explanation is, based on how well its premises fit the observed outcome (p.201, 204).

Prediction – Process by which AI (or humans) estimates the most probable outcome based on learned patterns (p.174–175).

Prediction & Forecasting – AI's ability to analyze past data to estimate future outcomes. Applications include finance, healthcare, meteorology, retail, education, sports, and urban planning (p.224–225).

Premises – The starting points of reasoning; can be facts, observations, predictions, preferences, or rules (p.195, 197–198).

Privacy – An ethical issue involving AI misuse of personal or sensitive data without consent (p.229).

Program – A set of written instructions a computer can execute to run an algorithm (p.139).

Qualitative data – Descriptive, categorical data that can be observed but not measured (e.g., gender, grade levels, types of cars). Must often be converted into quantitative form for visualization (p.18–19).

Quantitative data – Data that can be measured (continuous) or counted (discrete). Best visualized with line charts, scatter plots, or bar charts (p.16–17).

Queue – A linear data structure that follows FIFO. Items are added at the end and removed from the beginning (p.42–44).

Reasoning – A structured way of thinking to discover, decide, or explain. Outcomes may be inference, decision, or explanation (p.194, 206).

Reasoning and Decision-Making (AI) – AI's ability to evaluate options and select actions in a specific domain. Examples include recommender systems, logistics planning, dynamic pricing, and game-playing AIs (p.222–223).

Repetition – Strengthens associations and improves pattern recognition in both humans and AI (p.158–159, 166).

Root node – The topmost node in a tree structure from which all other nodes descend (p.46).

Scalability – How an algorithm's performance changes as dataset size increases. Sorting algorithms like insertion and bubble sort scale poorly due to quadratic growth (p.129–130, 137–138).

Scatter plot – A chart showing individual data points, often used to highlight relationships or correlations between two variables (p.14, 16–17).

Search index – Position within a dataset being checked during a search algorithm (p.71, 78).

Search range – Portion of a dataset still under consideration in a binary search; updated each iteration (p.78).

Searching – Fundamental operation required to locate a specific item in a dataset before modification, extraction, or processing (p.68, 83).

Sorting – The process of arranging data in a specific order based on a criterion, such as chronological order, increasing or decreasing value, or categories. Sorting makes it easier to search, interpret, and analyze data (p.13–14, 94–98).

Stack – A linear data structure that follows LIFO. Items can only be added or removed from the end (p.39–41).

Step count – The number of operations an algorithm performs. Used alongside execution time to measure performance (p.124, 126–127, 135).

Supervised learning – A type of machine learning that uses labeled data; the system is trained with examples

where the correct answer is provided (p.160).

Swap – Operation in bubble sort where two adjacent items exchange positions. Each swap requires three moves in memory (p.135).

Table – An arrangement of data in rows and columns; well-suited for organizing multiple sets of related data (p.12–13).

Termination – The final block of a program where a message indicates the task is complete and the program ends (p.110, 123).

Training – The process of improving a model using new data and corrections. It involves analyzing mistakes, refining rules, and updating the model to improve accuracy and generalization. Effective training depends on both the volume (amount) and variety (diversity) of training data (p.163–165, 178, 183–184).

Transparency and accountability – Ethical principle requiring AI decisions to be explainable and challengeable. Lack of transparency creates risks of unfair or harmful outcomes (p.229).

Tree – A non-linear hierarchical data structure with nodes connected along branches, including root, parent, and child nodes (p.31, 45–48).

Tuple – A linear data structure with fixed size and format, can store multiple data types together as a unit (p.37–38).

Uncertainty – The degree to which reasoning outcomes may not be definite, especially when based on predictions or incomplete premises (p.197, 204–205).

Underestimation/Overestimation – When predictions consistently fall below or above actual values (p.180–182).

Unsupervised learning – A type of machine learning that uses unlabeled data; the system identifies patterns and groups similar items without predefined labels (p.161).

Utility – A measure of how useful, valuable, or goal-fitting an outcome is; used to evaluate decisions (p.200).

Visualization – Presenting data in visual form (charts, graphs) to highlight patterns and trends (p.14–15).

Worst-case scenario – Situation where an algorithm must check all possible items before concluding, or in sorting, when data requires the maximum steps (e.g., reversed order for ascending sort). Represents the longest possible execution time (p.73–74, 124–127).

